# The galaxy-bias ladder under one-loop bias

A correlated bias prior across thin redshift bins,
under model A at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ ($z = 0$)

November 2026

## 1 Summary

**The question.** The forecast measures DESI galaxy clustering alone: the angular band powers $C_\ell^{gg}$ of thin redshift bins, computed without the Limber approximation, for six tracer groups (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four bin layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. How freely may the bias of each thin bin move within its group? This note answers with a ladder of four stated linear-bias priors, tests each against random bias histories, and carries the higher-order bias with the one-loop model A of `notes/higher_order_bias`. Linear bias alone is not enough: the companion note `notes/linear_bias_kmax` shows that a linear-bias fit to a one-loop truth at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ on the $\Delta d_{\rm com} = 62$ Mpc layout moves $\Omega_m$ by 2.0–2.4$\sigma$ and $(w_0, w_a)$ by 0.7–1.1$\sigma$ under every linear-bias prior. The same ladder with linear bias only ("v1" below, the notebook `EXAMPLE_EVALUATE_FISHER.ipynb`) set up the mechanics; model A runs on v1's kept band powers, so every difference from v1 is the bias model.

Symbols: $w(a) = w_0 + w_a(1 - a)$ is the dark-energy equation of state, with $a$ the scale factor. The figure of merit ${\rm FoM}(w_0, w_a) = [\det {\rm Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized $(w_0, w_a)$ error ellipse; larger is better. $\Omega_m$ is the matter density today in units of the critical density, $A_s$ the amplitude of the primordial curvature power spectrum (sampled as $10^9 A_s$), and $\sigma$ a parameter's marginalized one-sigma error. $D(z)$ is the linear growth factor, normalized to $D(0) = 1$.

**The scale cut.** $k_{\rm max}$ labels one cut per bin. Bin $i$ is cut at $k_{\rm max}/D(\bar{z}_i)$, with $\bar{z}_i$ its mean redshift: matter is less nonlinear at higher redshift, so the cut moves to higher wavenumber $k$ as $1/D$. The cut is the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, computed from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it. A band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; \alpha = 0.05, \tag{1}$$

and the kept bands are intersected with the shipped mask. The cut was built on the linear model. Every $k_{\rm max}$ in this note is the $z = 0$ value.

**The problem.** The default model ties all bins of a group to one amplitude, $b_{1,i} = B_g/D(z_i)$, with $b_{1,i}$ the linear bias of bin $i$, $z_i$ its redshift and $B_g$ one free amplitude per group $g$. This growth-tied model keeps the clustering amplitude $b_1 D$ of a group constant, as measured for

DESI-like luminous red galaxies [14]. The tying is what makes thin bins powerful, and it fails in two ways at once. It is too rigid: known physics moves each bin away from the group trend at the percent level (Appendix B). And it is inaccurate: a drift the model cannot express does not average away, it displaces the fitted cosmology, by many times the model's own error ellipse for the gentlest drift physics supplies, the slow debiasing of a conserved galaxy population (the accuracy test below). The opposite extreme, one free bias per bin, is accurate by construction but spends part of the data measuring the biases, 76 of them on $\Delta d_{\rm com} = 62$ Mpc. The model below sits between the two: bins may deviate from the group trend, but neighboring bins deviate together.

**The linear-bias model.** The linear bias of bin $i$ in group $g$ is

$$b_{1,i} \;=\; \frac{B_g}{D(z_i)}\,(1+\delta_i), \qquad \delta_i \;=\; s_g\,(z_i - \bar{z}_g) + \varepsilon_i, \tag{2}$$

with $B_g$ a free amplitude and $s_g$ a free tilt per group (flat priors), $\bar{z}_g$ the group's mean bin redshift, $\delta_i$ the fractional deviation from the group trend, and $\varepsilon_i$ the residual around the group's straight line. The residuals carry a Gaussian prior correlated within a group and zero across groups,

$$\langle \varepsilon_i\, \varepsilon_j \rangle \;=\; \sigma_g^2\, \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{3}$$

with $\sigma_g$ the prior width of one bin's residual in group $g$ and $L$ the correlation length in redshift. The tilt is free for two reasons: the shape $B_g/D$ is not the evolution of a conserved population, whose bias obeys $b - 1 \propto 1/D$ [9] (Appendix B.1), and every measured drift is close to a straight line across its group (Appendix C). In the Fisher matrix the prior becomes one projected precision block per group that charges only the residual's shape, never the level or the line (Appendix C.5; derived in Appendices I–M).

**What $\sigma_g$ and $L$ mean.** $\sigma_g$ ($\sigma_b$ in the appendices and figure panels, for a generic group) sets how far one bin may wander from its group's line. $L$ sets how far apart two bins must be before their residuals are independent: their correlation is $e^{-\Delta z/L}$ at a separation $\Delta z$, so bins much closer than $L$ deviate almost as one. Together they set the jump a bin may take relative to its neighbor, an rms of $\sigma_g\sqrt{2\,(1 - e^{-\Delta z/L})}$ (Appendix A.2). Physically, the line absorbs the coherent evolution of a selected sample, such as a flux threshold that rises with distance or the debiasing of a conserved population. The residual holds what a line cannot: the curvature of such a drift, and local features of the selection, such as the 4000 Å break crossing a filter edge (over $\Delta z \approx 0.03$–$0.05$) or the narrow redshift windows where sky lines hide the ELG [OII] doublet. A long $L$ describes a smooth residual, a short $L$ local features (Appendix C.4).

**The ladder.** The forecast tests four linear-bias priors, from rigid to free.

- **Growth-tied**: $b_{1,i} = B_g/D(z_i)$, one free amplitude per group and no tilt or residual; the default of the shipped yamls.
- **The baseline prior**: $\sigma_g = 3\%$ for BGS, 1% for the other five groups, and $L = 0.2$. The BGS width holds the curvature of its flux-limit drift: the $r < 19.5$ magnitude limit becomes a luminosity threshold that rises with redshift and drags the bias up by about

22% across the group, and a straight line removes all of that except a 2.4% rms curvature (Appendix E). The 1% of the other groups is a stated hedge for selection curvature not modeled here, about ten times the residual a conserved population leaves after its line (rms at most 0.16%, Appendix C.4). $L = 0.2$ is comparable to the group spans, so the residual is smooth across a group.

- **The pessimistic prior**: $\sigma_g = 3\%$ in every group and $L = 0.05$: local selection features in every group, as large as the largest known curvature, on the scale over which the 4000 Å break crosses a filter edge. That mechanism is known but its amplitude is not measured (Appendix B.4), so this prior grants it the largest measured size everywhere.
- **Free**: one $b_{1,i}$ per bin, each with a flat prior; the lower limit.

Why a ladder and not one prior? A prior is plausible if the bias histories physics gives us look like ordinary draws from it. That test is necessary, not sufficient, and it needs trusted histories; this note has two derived drift laws and no mock catalogs. So the forecast does not defend one prior. It first asks which priors recover the cosmology honestly, and quotes the FoM only for those: the pessimistic prior as the defensible headline, the baseline prior as what better knowledge of bias evolution would buy, and the free prior as the lower limit.

**What the two priors look like.** Figures 1 and 2 draw both priors around the known drifts. Under the baseline prior the neighbor residual rms at $\Delta z = 0.02$ is 0.44% (1.31% for BGS), the draws are smooth, and the known drifts look like ordinary members of the family; the curvature the BGS line leaves costs a $\chi^2$ per degree of freedom of about 0.8 under the residual prior on the $\Delta d_{\rm com} = 62$ Mpc bins. Under the pessimistic prior the neighbor rms is 2.44% in every group: the histories keep their lines but gain local features nobody has measured, which is the point of that prior.

**Model A.** Model A keeps this per-bin linear bias and adds the one-loop terms. As computed here, the real-space galaxy power spectrum of bin $i$ is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_0\, k^2 P_{\rm mm}(k) \;+\; \frac{\rm SN_0}{\bar n_i}, \tag{4}$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ holds the one-loop terms. These carry the quadratic bias $b_2$, the response of the galaxy density to the square of the matter density contrast, and the tidal bias $b_{s^2}$, its response to the squared tidal field; the third-order bias $b_3$ is zero. $\alpha_0$ is the $k^2$ counterterm, in $h^{-2}\,{\rm Mpc}^2$, which absorbs the sensitivity of the loops to small scales. $\rm SN_0$ is the white-noise power beyond Poisson, in units of the bin's Poisson power $1/\bar n_i$, with $\bar n_i$ the bin's number density. The higher-order biases follow each bin's own linear bias through calibrations, plus one offset per group:

$$b_{2,i} \;=\; b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} \;=\; -\tfrac{4}{7}\,(b_{1,i} - 1) + \Delta_{s,g}, \tag{5}$$

with $b_2^{\rm L}$ the separate-universe fit of Lazeyras et al. [3] and the second relation the co-evolution of a population formed with local bias [4]. Each group carries one $\alpha_0$ and one $\rm SN_0$. The priors are Gaussian and centered on zero, with widths 1 on $\Delta_2$, 0.5 on $\Delta_s$, $0.5/k_{\rm max}^2 = 88.9\,h^{-2}\,{\rm Mpc}^2$ on $\alpha_0$ (a 50% correction to the linear spectrum at $k_{\rm max}$, the rule of the DESI DR1 full-shape

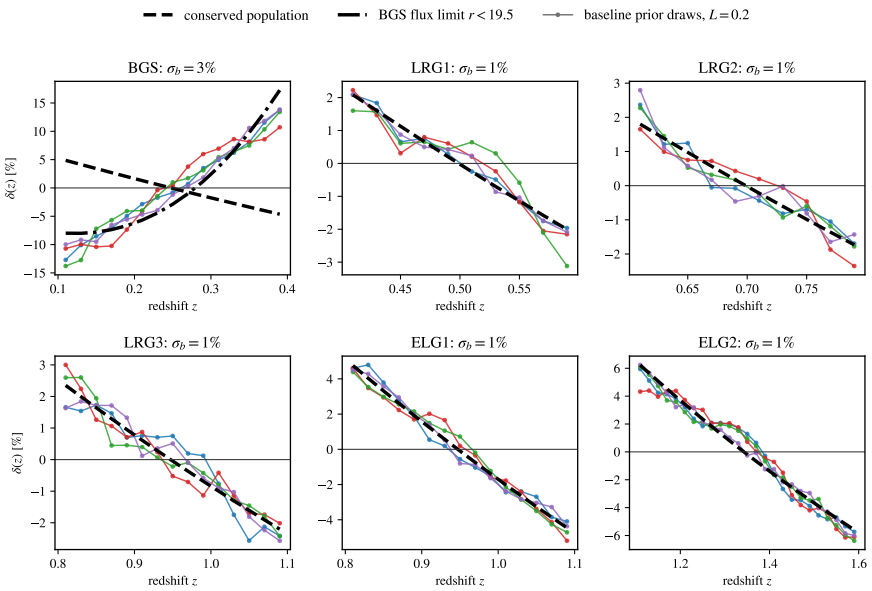



Figure 1: Draws of the baseline prior on the $\Delta z = 0.02$ bins, one panel per group ($\sigma_b$ in the panel title is the group's $\sigma_g$; $L = 0.2$). Thick dashed: the conserved-population drift. Thick dash-dot, on BGS: the $r < 19.5$ flux-limit drift. Thin curves with dots: four draws of the deviation $\delta(z)$, each on the line of the group's reference drift (the flux limit in BGS, the conserved population elsewhere), so draw and drift differ by the residual $\varepsilon$.

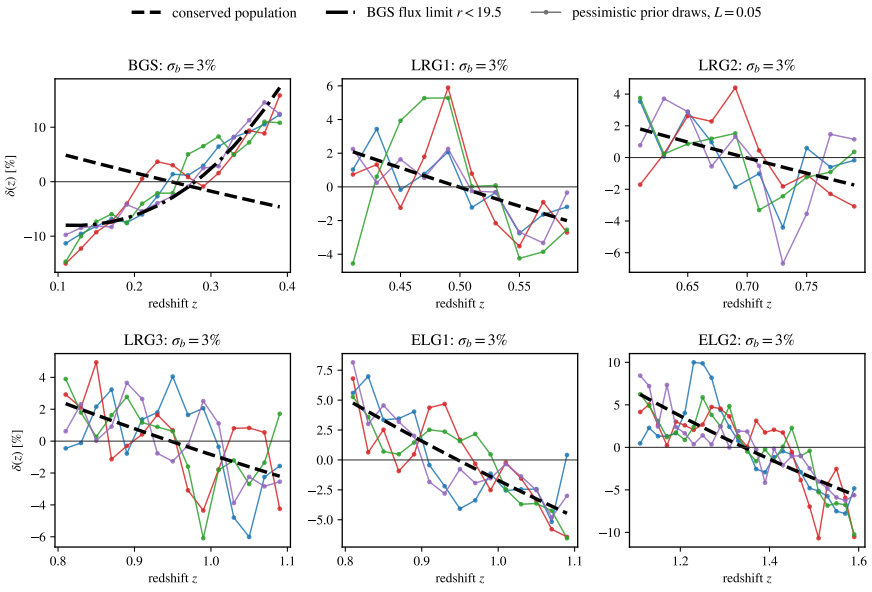



Figure 2: The pessimistic prior in the format of Figure 1: $\sigma_g = 3\%$ in every group and $L = 0.05$.

analysis [5, 6] moved to this cut), and 2 on $\mathrm{SN}_0$. Redshift-space distortions are linear (Kaiser), with each bin's redshift distribution convolved by the sampled velocity dispersion $\sigma_v$ (the Fingers-of-God smearing), as in v1.

Two variants of the higher-order sector sit on every linear-bias prior. **Model A baseline** has one $\Delta_2$, $\Delta_s$, $\alpha_0$ and $\mathrm{SN}_0$ per group with the priors above, 24 parameters. **Model A pessimistic** also tilts the $\Delta_2$, $\Delta_s$ and $\alpha_0$ of each group linearly in redshift, with flat priors on the tilts, up to 42 parameters; $\mathrm{SN}_0$ never tilts, because its Poisson part is known bin by bin.

**The accuracy test.** Each truth sets every bin's linear bias to its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere) plus a residual drawn from the baseline or the pessimistic prior (Appendix G.3). Under model A the truth's $b_2$ and $b_{s^2}$ follow its $b_1$ through Eq. (5). Every linear-bias prior fits every truth under both variants, centered on the growth-tied shape. Figure 3 draws the $(w_0, w_a)$ shift of each fit in units of its own ellipse, for 2000 such bias histories per case (v1's random seed), without data noise. Where a fit's prior equals the truth's scatter (the baseline prior in the top panel, the pessimistic prior in the middle one), the mean shift is not a bias: that fit's ellipse already covers this scatter. Its bottom panel shows the calibration drift discussed below.

For each parameter, the test asks one thing: does the fit's quoted error bar honestly cover where the fit actually lands? The verdict is

$$\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}, \qquad \chi^2 = \Delta p^{\mathsf{T}}\, \mathsf{C}_p^{-1}\, \Delta p, \tag{6}$$

with $\Delta p$ the recovered minus the true parameters, $\mathsf{C}_p$ the fit's marginalized covariance, $n_{\mathrm{dof}} = 2$ for $p = (w_0, w_a)$ and 1 for $\Omega_m$ or $A_s$, and $\langle \cdot \rangle$ the exact average over both sources of displacement: the data noise and the bias history (the known drift plus a random residual). A value of 1 means honest error bars; above 1 it is the factor by which they are too small, and below 1 they are conservative. It scores the recovery of the parameters, not the $\chi^2$ of the data: a biased fit can match the band powers well while its cosmology is wrong. Appendix G derives both statistics, and Figure 4 draws $\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}$ for model A baseline under the pessimistic truths.

Honest has a plain meaning. In many universes, each with its own noise realization and its own bias history, an honest $1\sigma$ bar covers the truth 68% of the time. For one parameter whose prior absorbs the known drift, a value $s$ of the statistic means a coverage of $2\Phi(1/s) - 1$, with $\Phi$ the cumulative distribution of a unit Gaussian: 68% at $s = 1$, 50% at 1.5, 38% at 2 (Appendix G.4).

The verdict (Figures 3 and 4) repeats v1's under model A. Numbers are for model A baseline unless stated.

- *The growth-tied prior fails under every truth and is discarded.* The conserved-population drift alone moves its $(w_0, w_a)$ by 7.9–11$\sigma$ (Appendix Q.1); its $(w_0, w_a)$ ellipse is 5.6–10.3 times too small (2.0–4.4 under model A pessimistic) and its $\Omega_m$ error bar 3.9–10.2 times. A model that does not fit the data has no meaningful FoM.

- *The baseline prior is honest only if nature's bias features are as smooth as it assumes*: $\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}} = 1.00$–$1.01$ under its own scatter. Under the rougher pessimistic scatter it fails, with $\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}} = 2.30$, 1.55 and 2.24 in $(w_0, w_a)$, 1.74, 1.29 and 2.66 in $\Omega_m$, and 1.58, 1.15 and 1.81 in $A_s$ on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$ and 0.05 (1.6–2.35 in $(w_0, w_a)$ under model A pessimistic). For one parameter the rms bias is at least $\sqrt{\langle \chi^2 \rangle - 1}$ in

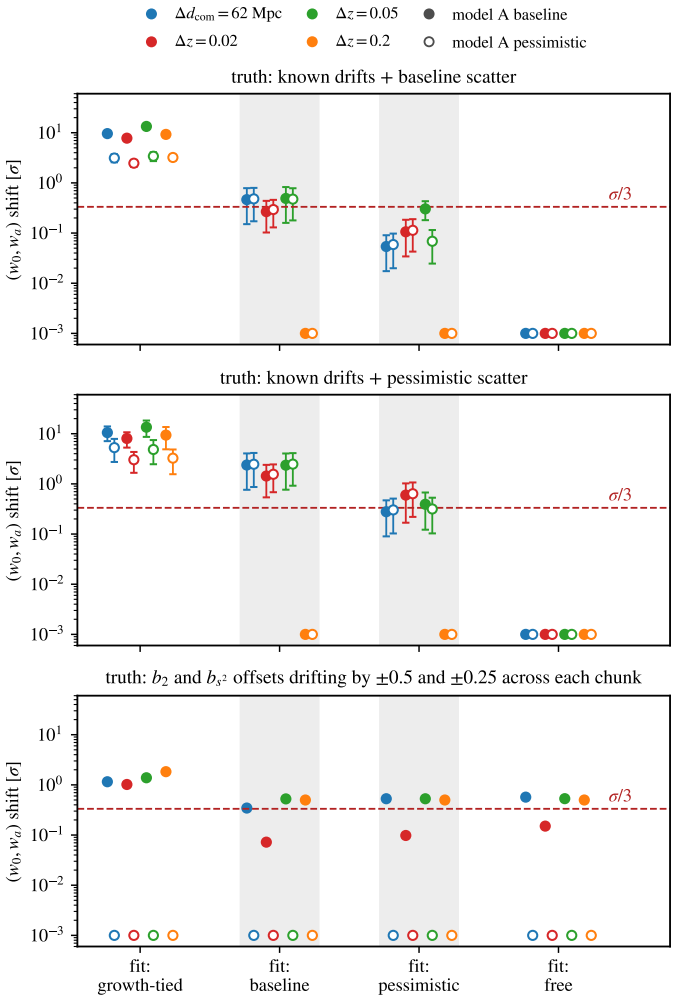



Figure 3: The accuracy test under model A: the $(w_0, w_a)$ shift of each fit in units of its own marginalized ellipse, under every linear-bias prior (horizontal axis). Filled: model A baseline; open: model A pessimistic; colors: the layouts. Top and middle: 2000 bias histories, the known drifts plus a residual from the baseline prior (top) or the pessimistic prior (middle), without data noise; points: the mean shift; bars: the 16th–84th percentiles. Bottom: the calibration-drift truth. Dashed line: $\sigma/3$. Gray columns: the baseline and pessimistic priors. Points at $10^{-3}$: exact zeros. $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

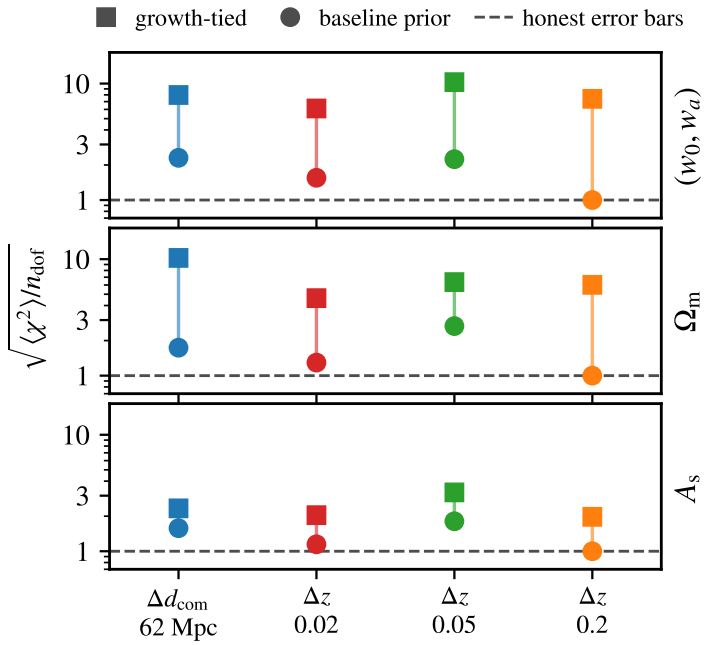
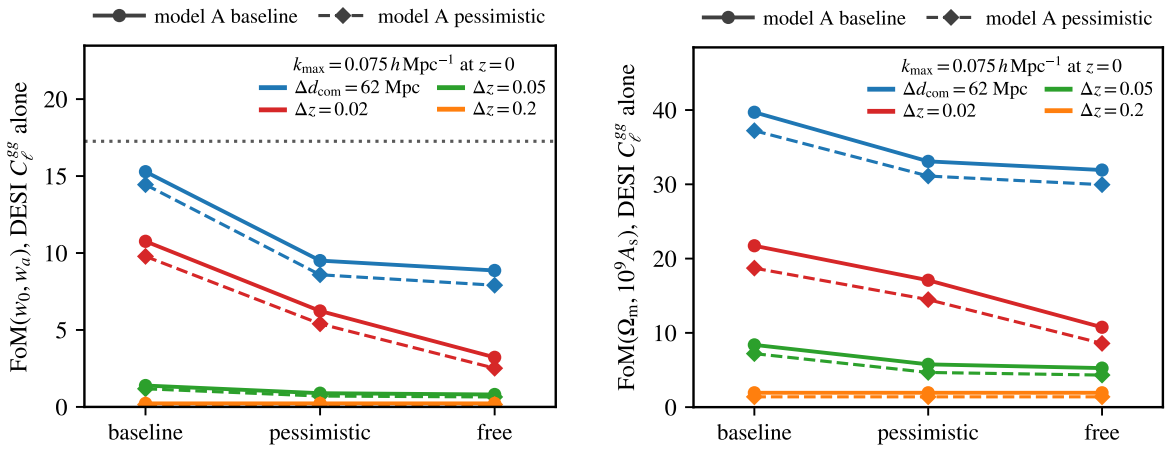



Figure 4: Does each fit's quoted error bar honestly cover where the fit lands? Plotted: $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, with $\chi^2 = \Delta p^{\mathsf{T}} \mathsf{C}_p^{-1} \Delta p$, $\Delta p$ the recovered minus the true parameters and $\mathsf{C}_p$ the fit's marginalized covariance, for $p = (w_0, w_a)$ ($n_{\rm dof} = 2$), $\Omega_m$ or $A_s$ ($n_{\rm dof} = 1$); $\langle\cdot\rangle$ averages over the data noise (from the band-power covariance) and the bias history (the known drifts plus a random residual from the pessimistic prior), under model A baseline. For one parameter, a value $s$ means that a fraction $2\Phi(1/s) - 1$ of these universes ($\Phi$ the unit-Gaussian cumulative distribution; at most that for growth-tied) has a $1\sigma$ bar that covers the truth: 68% at $s = 1$, the dashed line of honest error bars. Squares: the growth-tied prior; circles: the baseline prior. Omitted: the free prior (exactly 1); the pessimistic prior (at most 1.04); the baseline prior on $\Delta z = 0.2$ (1). $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

units of its error: about $1.4\sigma$, $0.8\sigma$ and $2.5\sigma$ in $\Omega_m$, whose $1\sigma$ bar covers the truth in only 44, 56 and 29% of universes instead of 68%. On $\Delta z = 0.2$ it is 1 only because each group holds one or two bins, so there the baseline prior is the free one.

- *The pessimistic prior is honest under both scatters*, at most 1.04 in every score, and conservative (0.85–1.00 in $(w_0, w_a)$) under baseline scatter. The free prior is exactly 1.

**The figure of merit.** A FoM means something only for a prior that passes the accuracy test: the pessimistic and free priors under both truths, the baseline prior only if bias features are as smooth as it assumes. Figure 5(a) gives FoM$(w_0, w_a)$ under both variants. It falls from the baseline prior to free under each variant; on $\Delta z = 0.2$ every group holds one or two bins, the free amplitude and tilt already free every bin, and the three priors coincide. Model A pessimistic on the pessimistic prior gives about 9, 5, 0.7 and 0.15 (layouts in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2), and model A baseline on the baseline prior about 15,




(a) (b)

Figure 5: **(a)** FoM$(w_0, w_a)$ and **(b)** FoM$(\Omega_m, 10^9 A_s)$ of DESI $C_\ell^{gg}$ alone under the baseline, pessimistic and free priors. Colors: the bin layouts. Solid lines: model A baseline; dashed lines: model A pessimistic. Dotted line in (a): the LSST Y1 3x2pt FoM$(w_0, w_a)$, 17.26. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

11, 1.4 and 0.2. On $\Delta d_{\mathrm{com}} = 62$ Mpc both sit below the LSST Y1 3x2pt reference of 17.26 [7].

Figure 5(b) gives the figure of merit of the matter density and the primordial amplitude, FoM$(\Omega_m, 10^9 A_s) = [\det \mathrm{Cov}(\Omega_m, 10^9 A_s)]^{-1/2}$ in the sampled units, from the same Fisher inverse. Under model A baseline it falls from the baseline prior to free from about 40 to 32 on $\Delta d_{\mathrm{com}} = 62$ Mpc, from 22 to 11 on $\Delta z = 0.02$ and from 8.4 to 5.3 on $\Delta z = 0.05$; on $\Delta z = 0.2$ it is about 1.9 under all three. Model A pessimistic on the pessimistic prior gives about 31, 14, 4.7 and 1.4. Appendix T sets these numbers next to v1's linear bias and to exact calibrations, neither of which is a result here.

**A drift of the calibrations.** The random truths move only the linear bias; the higher-order biases follow it through calibrations assumed exact up to one constant offset per group. A third truth tests that assumption. It drifts the offsets linearly with the bin redshift across each group, $\Delta_2$ from about $-0.5$ to $+0.5$ and $\Delta_s$ from about $-0.25$ to $+0.25$ (half the prior width at each end), and keeps $b_1$ at its fiducial. Under model A baseline, with the baseline, pessimistic or free prior, this drift shifts $(w_0, w_a)$ by 0.35–0.57$\sigma$ on $\Delta d_{\mathrm{com}} = 62$ Mpc and 0.50–0.53$\sigma$ on $\Delta z = 0.05$ and 0.2, all above $\sigma/3$, and 0.07–0.15$\sigma$ on $\Delta z = 0.02$ (bottom panel of Figure 3).

Model A pessimistic absorbs this drift exactly, by construction: its tilts have the drift's shape. The test shows only that the tilts work for a linear drift. A nonlinear drift of the calibrations within a group is untested. Protecting against the linear drift costs little on the thin layouts: model A pessimistic on the baseline prior gives about 14 and 10 against 15 and 11 for model A baseline.

**Recommendation.**

- **The headline**: model A pessimistic on the pessimistic prior, a FoM of about 9, 5, 0.7 and 0.15 on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2. It is the defensible forecast: honest under both families of random truths, and it absorbs a linear calibration drift exactly.

- **What better knowledge of bias evolution would buy**: model A baseline on the baseline prior, about 15, 11, 1.4 and 0.2. It holds only under two assumptions: each bin's linear-bias residual is no larger and no rougher than the baseline prior, and the calibrations of $b_2$ and $b_{s^2}$ hold within a group up to one constant offset. If the first fails at the pessimistic size, the $(w_0, w_a)$ ellipse is 1.55–2.3 times too small and the $\Omega_m$ bar covers the truth in only 29–56% of universes; if the second fails by the linear drift above, $(w_0, w_a)$ moves by about $0.5\sigma$ on three layouts.

The growth-tied prior is discarded under both variants, and the free prior under model A pessimistic (about 8, 2.5, 0.6 and 0.15) is the lower limit.

**Limits.** The covariance carries linear bias only, and its bias needs the update listed in the TODO below; comparisons between priors and variants hold it fixed. Shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ are linear responses; the truths are theory, not mocks, and only a linear calibration drift is tested. Two counterterm questions are open (Appendix S): its template draws more signal from beyond the cut than the cut allows the linear model, and at negative $\alpha_0$ the galaxy power turns negative within reach of a chain. Model B ($k_{\rm max} \approx 0.18\,h/$Mpc at $z = 0$, redshift-space loops) is not done. Appendix U lists every limitation.

# TODO

- **Update the DESI covariance.** Its bias must be the data vector's $b_i = B_g/D(\bar{z}_i)$; the covariance adapter (`covariance/lssty1xdesi_covariance.py`, function `_lens_bias`) currently passes the raw chunk amplitude $B_g$, which understates the high-redshift variances. Then regenerate the DESI covariances and rerun the FoMs: the absolute FoMs will drop by about 2.4–3.2 times.

- **Quote LSST Y1 3x2pt in Figure 5(b).** Compute the LSST Y1 3x2pt FoM$(\Omega_m, 10^9 A_s)$ from its Fisher matrix, with the same parameters and units as panel (b). Draw it in panel (b) as a horizontal dashed line, switch panel (a)'s LSST reference to the same dashed style so both panels match, and quote both values in the caption.

# A The models

This appendix first builds the prior without the tilt of Eq. (2): the kernel model, in which the exponential kernel describes the whole deviation $\delta_i$ of a bin from its group's trend. Appendix C adds the free tilt, and the kernel then describes the residual $\varepsilon_i$ around each group's line. In the appendices $b_i$ is the linear bias $b_{1,i}$ of bin $i$, and $\sigma_b$ is the prior width $\sigma_g$ of a generic group.

## A.1 Growth-tied, free, and correlated biases

The growth-tied model sets $b_i = B_g/D(z_i)$, with one free amplitude $B_g$ per group. The model multiplies the growth-tied bias by a fractional deviation $\delta_i$,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \tag{7}$$

with $B_g$ still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_b^2\,\exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{8}$$

and zero across groups. Here $\sigma_b$ is the prior standard deviation of a single bin's fractional deviation, and $L$ is the correlation length in redshift: bins much closer than $L$ deviate almost as one, bins much farther apart deviate independently.

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{9}$$

so that $b_i = \beta_i/D(z_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the $\beta_i$ within one group is

$$\mathsf{C} \;=\; s^2\,\mathsf{R}, \qquad s \;=\; \sigma_b\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{10}$$

The spread $s$ carries the units of a bias amplitude; the correlation matrix $\mathsf{R}$ is dimensionless, with ones on the diagonal.

## A.2 What $\sigma_b$ and $L$ mean

The kernel of Eq. (8) has two parameters, and they control two different things. The diagonal gives the wander of a single bin,

$$\mathrm{Var}(\delta_i) \;=\; \sigma_b^2, \tag{11}$$

so $\sigma_b$ sets how far any one bin may drift from the group's trend $B_g/D(z)$. The correlation coefficient $r$ of two bins $i$ and $j$ follows from the same kernel. By definition,

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; \frac{\sigma_b^2\,e^{-\Delta z/L}}{\sigma_b^2} \;=\; e^{-\Delta z/L}, \tag{12}$$

with $\Delta z = |z_i - z_j|$ the bin separation. The amplitude $\sigma_b$ cancels: the correlation is set by the separation and $L$ alone.

The jump between two bins now follows from the two pieces above: with $\mathrm{Var}(\delta_i) = \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \sigma_b^2\, e^{-\Delta z/L}$, the variance of the difference is $\mathrm{Var}(\delta_i) + \mathrm{Var}(\delta_j) - 2\,\mathrm{Cov}(\delta_i, \delta_j)$:

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_b^2 \left(1 - e^{-\Delta z/L}\right). \tag{13}$$

For separations much smaller than the correlation length, $\Delta z \ll L$, the exponential expands as $e^{-x} \approx 1 - x$, and

$$\mathrm{Var}(\delta_i - \delta_j) \;\approx\; \frac{2\,\sigma_b^2\,\Delta z}{L}, \qquad \text{rms jump} \;\approx\; \sigma_b \sqrt{\frac{2\,\Delta z}{L}}. \tag{14}$$

So $L$ sets the correlation between bins, $\sigma_b$ sets the wander of each bin, and *together* they set the jump a thin bin may take relative to its neighbor.

The combination that controls small separations is the roughness

$$\kappa \;\equiv\; \frac{\sigma_b}{\sqrt{L}}, \qquad \mathrm{Var}(\delta_i - \delta_j) \;\approx\; 2\,\kappa^2\,\Delta z. \tag{15}$$

The second form is the signature of diffusion: the variance of the increment grows linearly with the separation, exactly as the distance walked by a random walker grows linearly in time. On scales well below $L$ the deviation curve $\delta(z)$ therefore looks like a random walk with diffusion coefficient $\kappa^2$; only on scales beyond $L$ does the finite wander $\sigma_b$ pull it back toward zero.

**Worked example.** Take neighboring thin bins with $\Delta z = 0.02$ and correlation length $L = 0.05$. Then $\Delta z/L = 0.4$ and

$$r \;=\; e^{-0.4} \;=\; 0.670. \tag{16}$$

The exact rms jump from Eq. (13) is $\sigma_b\sqrt{2\,(1 - 0.670)} = 0.81\,\sigma_b$: for $\sigma_b = 10\%$ the neighboring bins may differ by 8.1% (one sigma), and for $\sigma_b = 1\%$ by 0.81%. The Taylor form of Eq. (14) gives instead $\sigma_b\sqrt{0.8} \;=\; 0.89\,\sigma_b$, an 8.9% jump. The approximation overestimates because $\Delta z/L = 0.4$ is not actually small: the expansion replaces $1 - e^{-x}$ by $x$, and $x$ exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about twenty percent here; the square root halves that, so the rms jump is overestimated by about ten percent.

### A.3 The free group amplitude and the projection

A plain Gaussian prior with covariance $\mathsf{C}$ would penalize moving all $\beta_i$ of a group up or down together, but that uniform shift is exactly the free amplitude $B_g$, which the growth-tied model never constrains. Keeping the penalty would make every comparison against the growth-tied model unfair. The repair removes the uniform direction from the prior: with $\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}$ the ones vector with $n$ entries ($n$ the number of bins in the group), the prior precision of a group's amplitudes is the projection

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{17}$$

Four statements, proven in the appendices, carry the construction. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing, so $B_g$ stays as free as in the growth-tied model. A pure shape change keeps exactly the penalty the plain prior gives it. $\mathsf{P}$ is positive semidefinite with

rank $n-1$: of a group's $n$ directions, one is free and the other $n-1$ are charged. And $\mathsf{P}$ is precisely the prior obtained by sampling $B_g$ with a flat prior and marginalizing it. Appendix I computes $\mathsf{P}$ for two bins, Appendix J derives these statements three ways, and Appendix K treats $n$ bins, including the growth-tied and free limits.

Across groups the prior of Eq. (8) is zero, so the full prior precision is block diagonal: one projected block per group, each group with its own free amplitude. How the blocks enter the Fisher matrix is Appendix N.

# B Which $L$ and $\sigma_b$ are physical

Which deviations from the growth-tied trend does known galaxy-bias physics produce? One case of bias evolution can be solved in closed form: a population whose members are conserved. This section derives that case, measures its deviation from the growth-tied trend, compares the deviation with what the prior permits, weighs the published measurements, and ends at the shape the measured drifts share.

## B.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes: no galaxy enters, none leaves. Assume the galaxies move with the matter on large scales (no velocity bias), and work in linear theory.

Under these assumptions the matter overdensity $\delta_m$ (the fractional fluctuation of the matter density) and the galaxy-count overdensity $\delta_g$ obey the same linear continuity equation, with the same peculiar velocity field on the right-hand side: number conservation gives each left-hand side, and the common motion gives the shared right-hand side. Subtracting the two equations, the excess clumping $\delta_g - \delta_m$ is frozen: whatever clumping difference the sample is born with never changes. Appendix D spells out every step.

Let $z_c$ be a reference redshift where the sample has linear bias $b_c$ (linear bias means $\delta_g = b\,\delta_m$), and let the matter grow as $\delta_m \propto D$. Carrying the frozen difference from $z_c$ to any other redshift gives [9]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}. \tag{18}$$

The excess clumping set when the galaxies formed stays fixed while the matter clustering grows underneath it, so the bias decays toward one with time. Tegmark and Peebles [10] extend the calculation to stochastic bias, where the conserved galaxy field traces the matter imperfectly (correlation $\rho_{gm} < 1$), and find the same decay of $b$ toward one; Appendix D summarizes their result.

The growth-tied shape $b = B/D$ freezes a different quantity. Constant $b\,D$ means $\delta_g = b\,\delta_m \propto (B/D)\,D = B$: the galaxy clustering amplitude itself does not grow. A conserved population never does that; its $\delta_g$ keeps growing with the matter piece $\delta_m(z)$ (Appendix D). Constant $b\,D$ instead describes a selection: at each redshift it picks a different and rarer set of objects, whose stronger clumping compensates the smaller $D$.

## B.2 The deviation from the growth-tied trend

Place the reference at a group's midpoint, $z_c$, with $b_c = B/D(z_c)$ from the group's fiducial amplitude $B$. The growth-tied bias is $b_{\rm gt}(z) = b_c\,D(z_c)/D(z)$ (gt for growth-tied) and the

conserved bias is Eq. (18). Their ratio defines the deviation $\delta(z)$ of Eq. (7), $b = b_{\rm gt}(1+\delta)$:

$$1+\delta(z) \;=\; \frac{1+(b_c-1)\,D(z_c)/D(z)}{b_c\,D(z_c)/D(z)} \;=\; \frac{D(z)/D(z_c)+b_c-1}{b_c}, \qquad \delta(z) \;=\; \frac{D(z)/D(z_c)-1}{b_c}. \tag{19}$$

At the midpoint $\delta = 0$. At lower redshift the matter has grown ($D$ larger than $D(z_c)$), so $\delta > 0$; at higher redshift $\delta < 0$. Because $D(z)$ is smooth, $\delta(z)$ is nearly linear in $z$ across a group.

In flat ΛCDM with $\Omega_m = 0.3$ and the project's fiducial amplitudes, Eq. (19) gives $\delta = +5.3\%$ and $-4.9\%$ at the ends of BGS ($B = 1.34$, $0.1 < z < 0.4$, $b_c = 1.5$), $+2.4\%$ and $-2.2\%$ at the ends of LRG1 ($B = 1.70$, $0.4 < z < 0.6$, $b_c = 2.2$), and $+2.6\%$ and $-2.3\%$ at the ends of LRG3 ($B = 1.70$, $0.8 < z < 1.1$, $b_c = 2.7$); across one $\Delta z = 0.05$ step at the midpoint, $\delta$ changes by 1.7%, 1.1% and 0.8%. In the language of the prior, a conserved population is a smooth tilt: coherent over the whole group, so $L$ longer than the group width, with an amplitude of a few percent at the group ends and a step of 0.3–0.7% between neighbors at $\Delta z = 0.02$.

Figure 6 draws these shapes. Look at any group: the dashed conserved curve is flatter than the solid $B_g/D$, because the frozen excess clumping is divided by a growing $D$, so the conserved bias rises more slowly toward high redshift. On BGS, the dash-dot flux-limit curve is the opposite: much steeper than $B_g/D$, and convex.

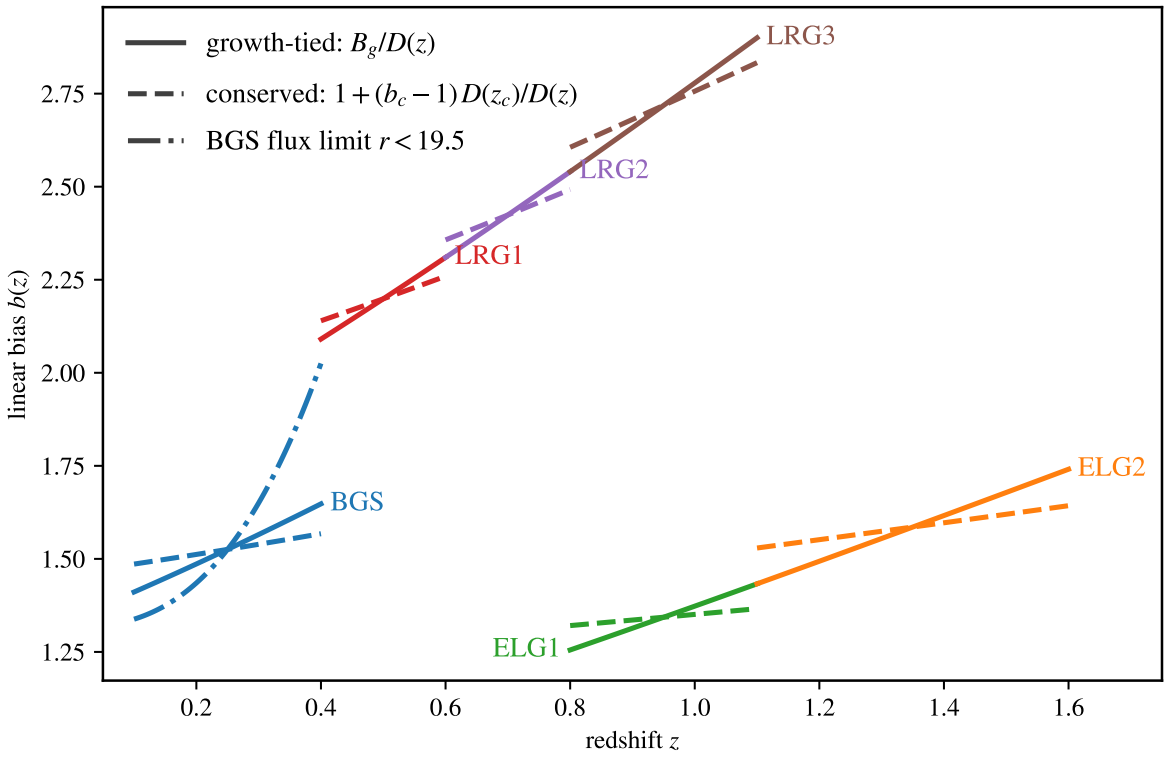



Figure 6: The bias $b(z)$ of every group. Solid: the growth-tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint, $b(z) = 1+(b_c-1)\,D(z_c)/D(z)$ (Eq. (18)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

### B.3 The prior's yardstick: the neighbor rms

The prior's quantity to compare with this tilt is the rms change between neighbors of Eq. (13), $\sigma_b\sqrt{2\,(1-e^{-\Delta z/L})}$. Over $\Delta z = 0.02$ (and over $\Delta z = 0.05$) it is 2.4% (3.4%) at $\sigma_b = 3\%$ and $L = 0.05$; 0.6% (0.9%) at $\sigma_b = 1\%$ and $L = 0.1$; 5.6% (6.8%) at $\sigma_b = 5\%$ and $L = 0.02$; 8.1% (11.2%) at $\sigma_b = 10\%$ and $L = 0.05$; and 22.5% (27.1%) at $\sigma_b = 20\%$ and $L = 0.02$. The conserved population of Appendix B.2 changes by 0.3–0.7% (0.8–1.7%).

At $\sigma_b = 1\%$ with $L = 0.1$ the prior allows neighbor differences of 0.6% over $\Delta z = 0.02$, matching the conserved drift. At $\sigma_b = 3\%$ with $L = 0.05$ the allowance is several times that drift, and at $\sigma_b = 10\%$ or 20% it reaches 8–27%, beyond any bias evolution identified in this note.

### B.4 What breaks conservation in a selected sample

Four mechanisms break the conserved-population assumptions.

**Threshold drift.** The target selection holds a roughly constant comoving density until the flux limit bites; past that redshift the sample keeps only its brighter members, which sit in more massive halos and are more biased. The resulting bias change is smooth in redshift, with $L$ comparable to the group width.

**Color-selection features.** The selection cuts in color space can respond quickly in redshift when a spectral feature crosses a filter edge. The sharpest common feature is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–$0.05$ at that break, and photometric scatter smears the transition further. Its amplitude in bias is unknown.

**Mergers.** Two members that merge become one, so the number of galaxies is not conserved even before any selection acts. The size of this effect on these samples is unknown.

**Emission-line selection.** An ELG enters the sample while its emission lines are strong, so the membership follows the star-formation phase of each galaxy rather than one conserved set of objects.

How much each of these moves the bias, and over which $\Delta z$, is not known from theory alone; it is a question about the measured selection function of each sample.

### B.5 What the measurements say

The mechanisms above can be weighed against published measurements. Each statement below carries a tag: *measured* means it is read from the cited source; *derived* means it is computed here from measured inputs; *assumed* means it is a judgment not yet backed by a measurement. Equation (18) itself is from Fry [9], which has no arXiv record; Tegmark and Peebles [10] (section "After the galaxy formation epoch") rederive it. Appendix F summarizes each paper below: the sample, the method and the numbers.

Massive red galaxies follow the conserved shape to the precision of the published tests. Analyses that follow one population of massive red galaxies across redshift find merger corrections of a few percent per Gyr (measured [11, 12, 13]), and the DESI-like LRG bias of [14] stays

within $\pm 2.3\%$ of the constant clustering amplitude $b = 1.5/D$ at a bin spacing of $\Delta z \approx 0.1$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high (measured biases; residuals derived here).

The selections drift. The LRG cuts hold the comoving density constant at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond (measured [15]); through the threshold-bias relation of [20] (measured), that flux limit makes the LRG3 bias drift by 6–11% per $\Delta z = 0.1$ on top of the $B/D$ shape (derived), matching the low end of the measured highest-bin offset. The magnitude-limited BGS (measured limit, [23]) drifts by 10–25% across its group through the same mechanism (derived).

The ELG bias shape is stable: HOD fits in two bins covering $0.8 < z < 1.6$ agree within one sigma (measured [26]). The one short-separation mechanism found in any tracer is the set of redshift windows where the [OII] 3727 Å doublet, which carries the ELG redshift, lands on sky emission lines and loses signal-to-noise, especially at $z > 1.5$ (mechanism measured [25]): each window is $\Delta z \approx 0.001$–$0.01$ wide, and its amplitude in bias is assumed, of order 1–2%: a judgment from the weak dependence of ELG bias on [OII] flux, not a measured bound.

None of the measurements collected here resolves bias below $\Delta z \approx 0.1$. The finest binning is the $\Delta z \approx 0.1$ of [14]; [26] use 0.3 and 0.5; and the lensing tomography of the same samples [17, 18, 19], built on the photometric samples of [16], uses about 0.15 with one free bias per bin. Below $\Delta z \approx 0.1$ the thin-bin statements of this note rest on the continuity equation and on the selection arguments.

### B.6 Drift versus loss of memory between bins

Every deviation found above, except the sky-line windows, is a monotonic drift: the conserved-population tilt, the LRG3 magnitude-limit drift and the BGS flux-limit drift each move in one direction across their group. A monotonic drift is a coherent, long-correlation deformation, a tilt; it is not bin-to-bin decorrelation, so it does not argue for a short $L$. Within the exponential kernel, the measured drifts point to a long $L$, and only the sky-line windows point to a short one. Appendix C gives the drifts to a free tilt instead.

## C The memory model

### C.1 Why a second model

The kernel model carries one amplitude and one length, and Appendix B.6 found that the measured deviations are not the kind it describes best. Every deviation found, except the narrow sky-line windows, is a line-like drift across its group: the BGS flux limit drifts by 10–25% across the group, the LRG3 magnitude limit by 6–11% per $\Delta z = 0.1$, and a conserved population by about 1% per $\Delta z = 0.05$ (Appendix B.5).

The kernel model can admit such a drift, but only at a price. To pass a 20% drift with $L \leq 0.1$ it needs $\sigma_b = 10$–$15\%$, and that spread also allows roughly 10% scatter between neighboring bins (Appendix B.3), which no identified mechanism produces. One $\sigma_b$ controls both the drift and the scatter, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two. The left panel of Figure 7 shows what the kernel model assumes a galaxy population does; Figure 8 shows what the measured physics does.

## C.2 Why the kernel forgets

The kernel's limitation is built into its correlation, $e^{-|z_i-z_j|/L}$, which decays with separation: at $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. Prediction makes the decay concrete. Appendix L shows that the best guess of the next bin given all earlier ones is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{20}$$

only the nearest neighbor matters, and the influence of any one bin on the rest of the chain fades as the product of the neighbor correlations, roughly $r^n$ after $n$ steps. The kernel forgets.

## C.3 The model, and why it remembers

The memory model adds one number per group: a tilt. The bias keeps the form of Eq. (7), and the deviation splits into a line and a residual,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; z_i - \bar{z}_g, \tag{21}$$

where $\bar{z}_g$ is the mean of the group's bin redshifts, $s_g$ is a free tilt per group with a flat prior (the same status as $B_g$), and the residual $\varepsilon_i$ carries the exponential kernel of Eq. (8), $\langle \varepsilon_i \varepsilon_j \rangle = \sigma_b^2\, e^{-|z_i-z_j|/L}$. This is the model of the Summary, Eqs. (2) and (3).

To see what the tilt changes, give the slope a Gaussian prior of variance $\tau^2$ first (the flat prior is the limit $\tau \to \infty$). The deviations then have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_b^2\, e^{-|z_i-z_j|/L}. \tag{22}$$

The new term does not decay with separation. Bins on the same side of the group center ($v_i v_j > 0$) are positively correlated, bins on opposite sides negatively. For separations much larger than $L$ the kernel term is gone and the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\big(\tau^2 v_i^2 + \sigma_b^2\big)\big(\tau^2 v_j^2 + \sigma_b^2\big)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_b : \tag{23}$$

the two ends of a group remember each other through the shared slope (Appendix M derives both steps).

Prediction shows the same memory. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{a} + \hat{s}_g\, v_{k+1} \;+\; r_k\,\big(\delta_k - \hat{a} - \hat{s}_g\, v_k\big), \tag{24}$$

where $(\hat{a}, \hat{s}_g)$ is the best-fitting straight line through all the past bins, weighted by $\mathsf{C}^{-1}$ (it needs at least two past bins). Every past bin now enters, through the fitted line; only the wiggle around the line stays Markov. Statistics calls Eq. (24) the universal-kriging predictor, prediction with an unknown linear mean; Appendix M proves it.

Figure 7 makes the difference visible on the BGS bins at $\Delta z = 0.02$. In the left panel, kernel draws at $\sigma_b = 10\%$, about what the kernel needs to admit a drift of BGS size, zigzag and forget: bins 0.1 apart are unrelated, and no draw looks like the thick flux-limit drift. In the right panel, memory draws with a tilt plus $\sigma_b = 2\%$ are lines with small wiggles, and the drift looks like a typical draw.

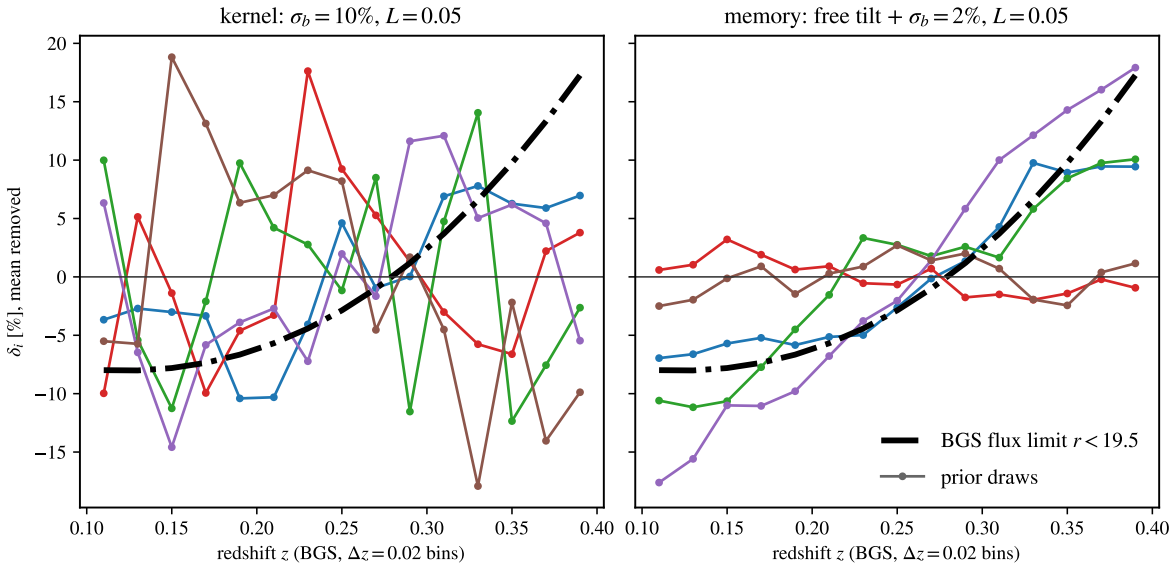



Figure 7: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed ($B_g$ is free). Thick dash-dot: the BGS flux-limit drift. Left: the kernel model, $\sigma_b = 10\%$ and $L = 0.05$. Right: the memory model, a tilt plus $\sigma_b = 2\%$ and $L = 0.05$; the tilt is free in the model and drawn here, for display only, from a Gaussian of width 0.6 per unit redshift.

### C.4 Why the memory model differs physically

A selected sample changes along a group in two physically different ways. The first kind is coherent: an evolution of the whole sample that moves every bin the same way. A flux or magnitude threshold that rises with distance, the debiasing of a conserved population, and the drift of the halo-mass threshold at fixed number density all act like this. The second kind is local: features of the selection that change bins independently and are forgotten over a short range, such as the sky-line windows and filter-edge crossings of Appendix B.4. The exponential kernel knows only the second kind, so it must inflate $\sigma_b$ to admit the first. The memory model gives the first kind to the tilt and the second to the kernel.

One straight line is enough for the coherent kind because the drifts are smooth functions of redshift, set by $D(z)$ and the distance modulus, which change on $\Delta z$ of order one. Over a group span of 0.2–0.5 the first-order Taylor term dominates. Fit the best straight line, with equal weights, through each group's conserved-population deviation of Eq. (19) at the $\Delta z = 0.05$ bin centers (derived; flat $\Lambda$CDM with $\Omega_m = 0.3$, amplitudes $B = 1.34$ for BGS, 1.70 for the LRGs and 0.84 for the ELGs, the reference at the group midpoint). The deviation spans 8.5% across BGS, 3.4%, 2.9% and 4.1% across LRG1–3, and 8.2% and 11.1% across ELG1 and ELG2, with slopes of $-16\%$ to $-34\%$ per unit redshift. The line leaves a residual of 0.02–0.16% rms, at most 0.27% (ELG2): a line removes the conserved-population drift to better than 0.3%.

The flux-limit drift is larger and not exactly straight. For BGS ($r < 19.5$, with $r$ the apparent-magnitude band rather than a correlation coefficient; no K-correction; the threshold-bias fit of [20] with $M_* = -20.44$ used without evolution; magnitudes in the $M - 5\log_{10} h$ convention that matches $M_*$; and the group amplitude set so the mean deviation vanishes),

the limiting magnitude runs from $-18.56$ to $-21.24$ between the first and last bin centers, the bias from 1.09 to 1.54, and the deviation spans 22.3% with a fitted slope of 89% per unit redshift. The line leaves 2.4% rms and 3.3% at most: the drift is convex, because the bias grows exponentially with the luminosity threshold (through the magnitude). So for BGS the memory model's residual scatter must be about 2–3%, or the model needs a quadratic term. Appendix E derives the flux-limit model step by step and lists what it leaves out: the threshold-bias fit frozen at $z \approx 0.1$, the missing K-correction and luminosity evolution, and the unmodeled fiber cut and BGS Faint sample.

Figure 8 carries both statements. In the top panel every conserved deviation is a straight line, and the thin gray fitted lines sit on top of them. In the bottom panel the conserved residuals are flat at zero, while the BGS flux limit leaves a $\pm 3\%$ U shape: that U is the curvature the memory model's $\sigma_b$ must absorb.

The two models use the same symbol $\sigma_b$ for different physical quantities. In the kernel model, $\sigma_b$ is the total departure of a bin from the $B_g/D$ trend: drift plus scatter. In the memory model, $\sigma_b$ is the departure from the group's own straight line: curvature plus local scatter. The physically plausible values differ accordingly: about 0.1% for a conserved population (the line fit above), 1–2% for the local selection features (Appendix B.5), and 2–3% for the BGS curvature.

The honest limit is unchanged: none of the measurements collected in Appendix B.5 resolves bias below $\Delta z \approx 0.1$. The memory model rests on the same evidence as the kernel model; it encodes that evidence in a form closer to the physics.

### C.5 What the prior charges

With the flat priors restored (slope variance $\tau^2 \to \infty$, and amplitude variance $W^2 \to \infty$ for $B_g$), the prior precision of a group's bin amplitudes becomes

$$\mathsf{P}_{\rm mem} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{25}$$

the same construction as Eq. (17) with one more column: $\mathsf{U}$ is the $n \times 2$ matrix whose columns are the ones vector and the centered redshifts $v$. The penalty it charges is

$$\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; \min_{a,\,s_g} \big(\delta - a\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - a\,\mathbf{1} - s_g\,v\big)\,: \tag{26}$$

the prior subtracts the best-fitting straight line, weighted by $\mathsf{C}^{-1}$, and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over the length $L$.

The structure of $\mathsf{P}_{\rm mem}$ separates local from global. $\mathsf{Q}$ is tridiagonal (Appendix K): a local object that links neighbors. The subtracted correction has rank two and is dense: it couples every bin to every other, because the fitted line listens to all of them. The kernel model already had one dense rank-one piece, the subtracted mean of Eq. (17); the memory model adds the slope. $\mathsf{P}_{\rm mem}$ annihilates both columns, $\mathsf{P}_{\rm mem}\mathsf{U} = 0$, is positive semidefinite, and has rank $n-2$ (Appendix M).

A group of one or two bins has $\mathsf{P}_{\rm mem} = 0$: an amplitude and a slope fix every bin, so the group is fully free. Among the four layouts this mostly affects $\Delta z = 0.2$, whose groups hold the fewest bins.

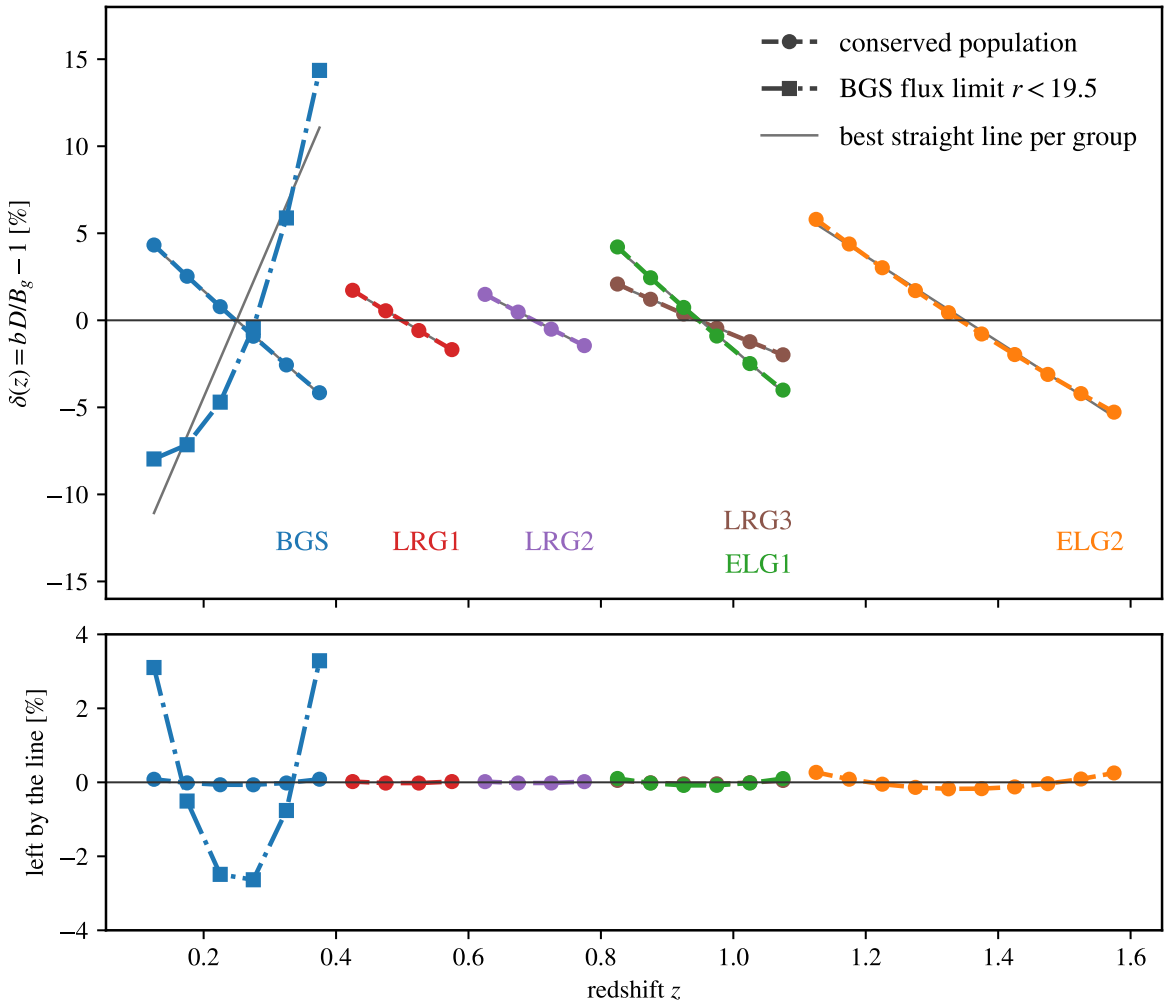



Figure 8: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with $B_g$ set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves; the conserved residuals are under 0.3%, the BGS flux-limit residual is convex, about $\pm 3\%$.

### C.6 What the slope costs

The tilt has a physical price. In linear theory the clustering band powers of bin $i$ scale as $b_i^2\,D(z_i)^2$, so the data measure the product $b\,D$, not the factors. Why a free tilt costs growth information takes one line: with $f = d\ln D/d\ln a$ the linear growth rate, $d\ln D/dz = -f(z)/(1+z)$, so inside a group

$$\frac{d\ln(bD)}{dz} \;=\; \frac{d\ln b}{dz} \;-\; \frac{f(z)}{1+z}. \tag{27}$$

A free tilt in $b$ removes the channel that measures $f$ through the amplitude evolution inside a group. The free amplitude $B_g$ already handed each group's overall growth to the bias; the

free tilt also hands over the relative growth inside a group.

Not everything is lost to the degeneracy. The redshift-space terms scale as $b\,f$, not as $b^2$, and measure $f$ directly; the geometric distances and the shape of the power spectrum are untouched; and the comparison of growth between groups survives, because each group gives away only its own interior tilt, a statement this note does not test. This is why a growth-tied model plus a free tilt sits below the growth-tied model, and why it should sit well above the free one. The limit $\sigma_b \to 0$ of the memory model isolates the question: it measures how much of the figure of merit came from growth measured inside groups.

### C.7 The limits

The memory model contains the same two endpoints as the kernel model, shifted by the tilt. As $\sigma_b \to 0$ the residuals vanish and every group is a rigid line with a free level and a free slope: the growth-tied model plus a free tilt per group. As $\sigma_b \to \infty$ the prior charges nothing and the biases are free. Between the endpoints, $\sigma_b$ and $L$ keep the same meanings as before, now applied to the wiggles around each group's line.

### C.8 The two models in one paragraph

The intermediate bias model frees one amplitude $B_g$ per tracer group and lets each bin deviate fractionally from the group's constant-amplitude trend $B_g/D(z)$, with deviations correlated over a redshift length $L$ and a per-bin wander $\sigma_b$. In the Fisher matrix this becomes one projected precision $\mathsf{P}$ per group, added to the bin-amplitude block: $\mathsf{P}$ charges for differences between neighboring amplitudes but never for a uniform shift of the group. The unprojected inverse covariance links only neighboring bins directly (it is tridiagonal, even though the correlation itself reaches far); the projection adds one dense rank-one piece, the subtracted mean. The prior reduces to the growth-tied model as $L \to \infty$ and to free flat-prior biases as $\sigma_b \to \infty$. The memory model adds a free tilt per group and reserves $\sigma_b$ for the residuals around each group's line, so it admits the measured drifts with $\sigma_b$ of about 1–3%, where the kernel alone needs about 10%.

## D The bias continuity equation, step by step

The matter density $\rho_m(\mathbf{x}, t)$ defines the matter overdensity $\delta_m = (\rho_m - \bar{\rho}_m)/\bar{\rho}_m$, with $\bar{\rho}_m(t)$ the mean density; the galaxy number density $n_g$ defines the galaxy-count overdensity $\delta_g$ the same way. Conservation of mass and of galaxy number gives each field a continuity equation. In comoving coordinates, keeping only terms linear in the fluctuations (the convective term $\mathbf{v} \cdot \nabla\delta$ is second order and dropped),

$$\frac{\partial \delta_m}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla \cdot \mathbf{v}_m, \qquad \frac{\partial \delta_g}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla \cdot \mathbf{v}_g, \tag{28}$$

where $t$ is cosmic time, $a$ the scale factor, and $\mathbf{v}_m$ and $\mathbf{v}_g$ the peculiar velocity fields of the matter and of the galaxies.

The two velocity fields are the same. On large scales the galaxies carry no pressure of their own and fall in the same gravitational potential as the matter, so a conserved sample rides the matter flow: $\mathbf{v}_g = \mathbf{v}_m$. This is the no-velocity-bias assumption; Tegmark and Peebles [10] state it as galaxies behaving as test particles that fairly trace the large-scale velocity field.

With equal right-hand sides, subtracting the two equations of Eq. (28) leaves

$$\frac{\partial}{\partial t}\big(\delta_g - \delta_m\big) \;=\; 0 \quad\Longrightarrow\quad \delta_g(z) - \delta_m(z) \;=\; \delta_g(z_c) - \delta_m(z_c) \tag{29}$$

after integrating in time from the reference redshift $z_c$ to any other redshift $z$: the excess clumping keeps its initial value forever.

Now substitute the bias. At $z_c$ the sample has linear bias $b_c$, so $\delta_g(z_c) = b_c\,\delta_m(z_c)$ and the frozen difference is $(b_c - 1)\,\delta_m(z_c)$. The matter grows as $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$, with $D$ the linear growth factor. Then

$$\delta_g(z) \;=\; \delta_m(z) + \big(b_c - 1\big)\,\delta_m(z_c) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{30}$$

and dividing by $\delta_m(z)$ gives the bias $b(z) = \delta_g/\delta_m$ of Eq. (18). Dividing instead by the growth-tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (19), $\delta(z) = (D(z)/D(z_c) - 1)/b_c$: positive at redshifts below $z_c$, where the matter has grown past its reference value, and negative above.

The debiasing limit follows from Eq. (18) alone. As $D$ grows, the frozen term is divided by an ever larger number, so $b \to 1$: from above when $b_c > 1$ and from below when $b_c < 1$. Gravity debiases a conserved sample, because the part of $\delta_g$ that traces the matter keeps growing while the frozen excess does not.

Tegmark and Peebles drop the assumption that $\delta_g$ is proportional to $\delta_m$ point by point. They describe the pair of fields by three numbers: the matter fluctuation amplitude $\sigma$, the bias $b$ (the ratio of the galaxy to the matter fluctuation amplitude) and the correlation coefficient $\rho_{gm}$ between the two fields; $\rho_{gm} < 1$ is stochastic bias, the galaxy field tracing the matter imperfectly. (The subscript keeps it apart from the neighbor correlation $r$ of Eq. (12).) The same continuity argument evolves all three (their section "After the galaxy formation epoch"): with $D$ normalized to one today and the subscript zero marking today,

$$b_0 = \big[(1-D)^2 + 2D(1-D)\,b\,\rho_{gm} + D^2 b^2\big]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\,b\,\rho_{gm}}{b_0}, \tag{31}$$

where $b$ and $\rho_{gm}$ are the values at the epoch whose growth factor is $D$.

Both $b_0$ and $\rho_{gm,0}$ approach one as clustering grows, whatever their initial values: the matter-sourced part of the galaxy field grows with $D$ while the frozen part does not, so the field is progressively dominated by the piece that traces the matter. Perfect correlation is a fixed point: for $\rho_{gm} = 1$, Eq. (31) collapses to $b_0 - 1 = (b-1)\,D$, which is Eq. (18) again. The conserved population of this note is the perfectly correlated special case of their model.

Each assumption fails in a specific way for a DESI sample. Membership is not conserved: the flux limit and color cuts admit and remove galaxies as redshift grows (the threshold drift and color-selection features of Appendix B.4), mergers turn two members into one, and an ELG belongs to the sample only during its star-forming phase. The no-velocity-bias and linear-theory assumptions hold well on the scales this forecast keeps; conservation is the assumption that breaks.

The script of Appendix O integrates Eq. (28) numerically, with $\delta_m$ from the linear growth equation, and recovers the closed form of Eq. (18) to a relative accuracy of $2 \times 10^{-9}$.

# E The BGS flux-limit bias model, step by step

Appendix C.4 quotes a BGS bias drift driven by the survey's flux limit: a 22.3% span of the deviation, a convex shape, and a 2.4% rms residual around the best straight line. This appendix derives that model in four steps, tabulates every intermediate quantity, explains the shape, and then lists what the model leaves out. The model lives in the function `bgs_flux_limit_bias` of `plots/make_theory_data.py`, and the script of Appendix O checks the numbers below.

## E.1 The four steps

**Step 1: the flux limit becomes a luminosity threshold.** BGS Bright keeps every galaxy brighter than apparent magnitude $r = 19.5$ (measured limit, [23]). Apparent magnitude $m$ and absolute magnitude $M$ are related by $m = M + 25 + 5\log_{10}[d_L/(h^{-1}\,\mathrm{Mpc})] + K(z)$, where $d_L$ is the luminosity distance, $h$ is the Hubble constant in units of 100 km/s/Mpc, and $K(z)$ is the K-correction: the magnitude shift that accounts for a fixed filter sampling a redshifted spectrum [22]. The model sets $K = 0$. The faintest absolute magnitude that passes the cut at redshift $z$ is then

$$M_{\rm lim}(z) - 5\log_{10} h \;=\; 19.5 - 25 - 5\log_{10}\frac{d_L(z)}{h^{-1}\,\mathrm{Mpc}}, \qquad d_L(z) \;=\; (1+z)\,\chi(z), \tag{32}$$

with $\chi(z)$ the comoving distance of flat ΛCDM with $\Omega_m = 0.3$, the background of these appendices. Every magnitude in this appendix carries the $M - 5\log_{10} h$ convention, the convention of the $M_*$ in step 3. Brighter means more negative: between the first and last $\Delta z = 0.05$ bin centers the threshold brightens by 2.68 magnitudes (Table 1).

**Step 2: the sample at each redshift is a luminosity-threshold sample.** At redshift $z$ the model sample holds every galaxy brighter than the threshold, $M < M_{\rm lim}(z)$. Each thin bin is therefore a luminosity-threshold sample, and the threshold brightens with distance: the one selection function $r < 19.5$ turns into a different threshold sample in every bin.

**Step 3: the threshold sets the bias through the Zehavi et al. fit.** Zehavi et al. [20] measured the large-scale bias of luminosity-threshold samples in the SDSS main sample at $z < 0.25$ (effective redshift about 0.1) and fit it with

$$b(>L) \;=\; 1.06 + 0.21\,(L/L_*)^{1.12}, \qquad L/L_* \;=\; 10^{-0.4\,(M_{\rm lim} - M_*)}, \tag{33}$$

where $L$ is the threshold luminosity of the sample, and $L_*$ and $M_* - 5\log_{10} h = -20.44$ are the characteristic luminosity and magnitude of the SDSS $r$-band luminosity function [21]. Appendix F summarizes the measurement itself. The model applies the fit at every redshift of the group, without evolution; it also reads the published left side $b\,(\sigma_8/0.8)$ as $b$, with $\sigma_8$ the rms matter fluctuation in spheres of $8\,\mathrm{Mpc}/h$.

**Step 4: the deviation from the growth-tied shape.** The deviation $\delta$ of Eq. (7) compares this bias with the growth-tied shape $B_g/D(z)$, where $D$ is the linear growth factor normalized to $D(0) = 1$. Because the group amplitude $B_g$ is free, only the shape of $b(z)$ matters, and

the model fixes the amplitude $B$ by making the mean deviation over the group's bin centers vanish:

$$\delta(z_i) \;=\; \frac{b(z_i)\,D(z_i)}{B} - 1, \qquad B \;=\; \frac{1}{n}\sum_{j=1}^{n} b(z_j)\,D(z_j), \tag{34}$$

with $z_1, \ldots, z_n$ the bin centers of the group.

Table 1 evaluates the chain at the $\Delta z = 0.05$ bin centers of the BGS group, the same evaluation points as Figure 8 and the numbers of Appendix C.4: the threshold runs from $-18.56$ to $-21.24$, the bias from 1.09 to 1.54, and the deviation spans 22.3%. On a coarser grid the endpoint values move; at $z = 0.11$ and $z = 0.39$, for example, the bias runs from 1.08 to 1.59.

| $z$ | $d_L$ [$h^{-1}$ Mpc] | $M_{\rm lim} - 5\log_{10} h$ | $L/L_*$ | $b$ | $D$ | $\delta$ |
|---|---|---|---|---|---|---|
| 0.125 | 410 | $-18.56$ | 0.18 | 1.09 | 0.937 | $-8.0\%$ |
| 0.175 | 591 | $-19.36$ | 0.37 | 1.13 | 0.913 | $-7.2\%$ |
| 0.225 | 783 | $-19.97$ | 0.65 | 1.19 | 0.890 | $-4.7\%$ |
| 0.275 | 984 | $-20.46$ | 1.02 | 1.27 | 0.867 | $-0.4\%$ |
| 0.325 | 1192 | $-20.88$ | 1.50 | 1.39 | 0.845 | $+5.9\%$ |
| 0.375 | 1409 | $-21.24$ | 2.10 | 1.54 | 0.824 | $+14.4\%$ |

Table 1: The BGS flux-limit chain at the $\Delta z = 0.05$ bin centers: the luminosity distance $d_L$ (Eq. (32)), the threshold magnitude $M_{\rm lim}$, the threshold luminosity $L/L_*$ and bias $b$ (Eq. (33)), the growth factor $D$, and the deviation $\delta$ from the growth-tied shape (Eq. (34)).

### E.2 Why the drift is convex

The two factors of $b\,D$ change at very different rates: across the bin centers of Table 1 the bias rises by 41% while $D$ falls by 12%, so the deviation inherits the shape of the bias curve. That curve is flat at low redshift and steep at high redshift for one reason: the luminosity ratio is exponential in the threshold magnitude, $L/L_* = 10^{-0.4\,(M_{\rm lim} - M_*)}$. Below $L_*$ the term $0.21\,(L/L_*)^{1.12}$ is small against the constant 1.06 (0.03 at the first bin center), so the bias stays near the faint-end floor of the fit. The threshold crosses $L_*$ near $z = 0.27$, and past the crossing the same term dominates the growth of $b$ and rises exponentially with the threshold magnitude (0.48 at the last center). A curve that is flat on one side of a crossing and exponential on the other is convex across the group.

A straight line through a convex curve leaves a U-shaped residual. Quantitatively, the deviation at the six centers is almost exactly a parabola: adding a quadratic term cuts the residual from 2.4% to 0.1% rms. The fitted quadratic coefficient is 3.8 per unit redshift squared, and a pure parabola with that coefficient, sampled at the six centers with its best line removed, leaves residuals of $+3.2\%$ at the ends and $-2.6\%$ in the middle, rms 2.4%: the measured residuals $(+3.1, -0.5, -2.5, -2.6, -0.8, +3.3)\%$ of Figure 8 are this U shape. The 2.4% rms of Appendix C.4 is therefore the curvature of the drift, the one part of it a free tilt cannot absorb.

### E.3 What the model leaves out

Each omission below states the direction in which it pushes the drift. None of them changes the qualitative shape: a smooth, monotonic, convex rise of the bias toward the far edge of the

group.

**The threshold-bias relation is frozen at $z \approx 0.1$.** Equation (33) is a fit to clustering at $z \lesssim 0.25$, applied here up to $z = 0.4$. Two neglected evolutions pull in opposite directions. First, the bias at fixed luminosity grows with redshift: for a conserved population the continuity equation of Appendix D gives $b - 1 \propto 1/D$ (Eq. (18)), so a threshold sample at $z = 0.4$ is more biased than the $z \approx 0.1$ calibration reports. This steepens the drift. Second, galaxies were brighter in the past: Blanton et al. [21] measure the luminosity evolution of the SDSS $r$-band luminosity function as a brightening of $M_*$ by $Q \approx 1$–$2$ magnitudes per unit redshift. A fixed threshold therefore sits lower relative to $L_*(z)$ at high redshift, which lowers $L/L_*$ and the bias there. This flattens the drift. The flattening is a fraction of the effect, not all of it: $Q\,(z - 0.1)$ reaches 0.3–0.6 magnitudes at the last bin center, against the 2.68 magnitudes the threshold itself brightens across the group.

**No K-correction.** The model drops the $K(z)$ of step 1. Two pieces are dropped with it. The bandpass-shift part alone, the special case of observing a spectrum flat in wavelength through its own filter, is $2.5 \log_{10}(1 + z)$, which is 0.37 magnitudes at $z = 0.4$ [22]; a red spectrum, falling toward the blue, makes the $r$-band correction larger still. In addition, the $M_*$ of step 3 is measured in the $r$ band shifted to $z = 0.1$ [21], while Eq. (32) produces an observed-band magnitude; the K-correction is exactly the bridge between the two bands, and the model omits the bridge. A positive $K$ makes the true threshold brighter at high redshift than Eq. (32) says, so the true $L/L_*$ and bias are higher there: the omission steepens the drift.

**Only $r < 19.5$ is modeled.** BGS Bright also imposes a fiber-magnitude cut for redshift success, and BGS Faint, the color-selected extension at $19.5 < r < 20.175$, is absent from the model [23]. A sample that includes BGS Faint has a limit 0.675 magnitudes fainter at every redshift, which lowers $L/L_*$ and the bias everywhere and moves the $L_*$ crossing to higher redshift: a flatter drift. The fiber cut and the color selection change the galaxy mix in a way the threshold fit of Eq. (33) cannot follow, and this model assigns them no direction.

**Scope.** The model is an order-of-magnitude estimate. Its reliable content is the shape of the drift (smooth, monotonic, convex) and its size to a factor of order unity, and that is the content this note uses: Appendix C.4 sizes the memory model's residual scatter for BGS from the 2–3% curvature, and the baseline prior of the Summary takes $\sigma_g = 3\%$ for BGS from it. It is not a prediction of the measured DESI BGS bias. The upgrade path replaces Eq. (33) with DESI's own measurement: the One-Percent Survey analysis of Prada et al. [24] fits the BGS clustering as a function of absolute magnitude and redshift with abundance matching, and reports the dependence of the large-scale bias on absolute magnitude; that relation, plus the measured K-corrections and luminosity evolution above [22, 21], would replace steps 1 and 3 with calibrated inputs.

# F What the literature says, paper by paper

This appendix summarizes each measurement behind Appendix B.5, for a reader who has not read the papers. The tags are those of that section: measured, derived, assumed.

### F.1 Conserved-population tests of massive red galaxies

**Wake et al. [11].** What was measured: the two-point correlation function of over 10,000 luminous red galaxies (LRGs) from SDSS and the 2SLAQ survey at $0.19 < z < 0.55$, matched in rest-frame color and luminosity, with stellar masses above $10^{11}\,h^{-1}M_{\odot}$. Method: the redshift-space and real-space autocorrelation of the matched samples, interpreted with a halo model. Result: no significant evolution of the correlation amplitude, but a strictly merger-free conserved population is rejected; a merger rate of $7.5 \pm 2.3\%$ between $z = 0.55$ and $z = 0.19$, about 2.4% per Gyr, fits the data. Implication here: LRG deviations from conserved evolution are a few percent in number per Gyr (measured), and the induced bias change is smaller, because merging moves galaxies between halos of similar mass.

**Tojeiro et al. [12].** What was measured: the merger rate and clustering evolution of 250,000 BOSS galaxies up to $z \approx 0.7$, connected to the SDSS-I/II LRGs at lower redshift. Method: each BOSS galaxy receives a likelihood of being a progenitor of the LRG sample, built from the fossil record of the LRGs (star-formation and metallicity histories); the weighted sample's large-scale clustering is compared between the surveys. Result: the sample evolves by merging at $2 \pm 1.5\%$ per Gyr, and the weighted clustering evolution is consistent with the continuity-equation prediction for a conserved population that moves with the matter. Implication here: the conserved shape holds to the percent-per-Gyr level for these galaxies (measured).

**Guo et al. [13].** What was measured: the projected two-point correlation function of about 260,000 CMASS galaxies at $0.43 < z < 0.7$ (BOSS DR9). Method: subsamples of fixed comoving number density built with redshift-dependent magnitude thresholds, so that the same population is followed across redshift. Result: the clustering of the fixed-density samples is consistent with a conserved population evolving under gravity alone, and red-sequence clustering at fixed luminosity shows no significant evolution. Implication here: the same (measured).

### F.2 The DESI luminous red galaxies

**Zhou et al. [14].** What was measured: the redshift-dependent clustering of 2.7 million DESI-like LRGs over 5655 square degrees at $0.4 < z < 0.9$, in photometric-redshift bins of width $\Delta z \approx 0.1$. Method: projected correlation functions with photometric redshifts of 0.02 precision, fit with a five-parameter halo occupation distribution (HOD). Result: the HOD parameters show little evolution except in the highest bin, and the bias follows a constant clustering amplitude; Table 2 lists the measured biases and their residuals around $b = 1.5/D$ (derived here): within $\pm 2.3\%$ below $z = 0.83$, and $+6.4 \pm 6.4\%$ at $0.84 < z < 0.93$. Implication: LRG deviations at $\Delta z \approx 0.1$ spacing are at most a few percent below $z = 0.83$ (measured plus derived).

**Zhou et al. [15].** What was measured: the DESI LRG target selection and its spectroscopic performance. Method: selection cuts in $g$, $r$, $z$ and $W1$ (the WISE 3.4 micron band), validated on Survey Validation spectra. Result: a sliding color-magnitude cut holds the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$; the overall faint limit is $z_{\mathrm{fiber}} < 21.6$, where $z_{\mathrm{fiber}}$ is the $z$-band magnitude expected within one DESI fiber; 98.9% of observed targets yield confident redshifts. Implication: inside LRG1 and LRG2 the selection gives the threshold no room to

| photometric bin | $b$ | residual around $1.5/D$ |
|---|---|---|
| 0.41–0.50 | $1.89 \pm 0.03$ | $-0.4\% \pm 1.6\%$ |
| 0.50–0.61 | $2.04 \pm 0.02$ | $+2.3\% \pm 1.0\%$ |
| 0.61–0.72 | $2.07 \pm 0.03$ | $-1.6\% \pm 1.4\%$ |
| 0.75–0.83 | $2.24^{+0.07}_{-0.08}$ | $+0.4\% \pm 3.4\%$ |
| 0.84–0.93 | $2.48 \pm 0.15$ | $+6.4\% \pm 6.4\%$ |

Table 2: The DESI-like LRG bias of Zhou et al. [14] (measured) and its residual around the constant clustering amplitude $b = 1.5/D$ (derived, with flat ΛCDM $\Omega_m = 0.3$ growth factors at the bin centers).

jump (measured); past $z \approx 0.8$ the fixed flux limit drives the LRG3 drift of Appendix B.5 (derived). The chain of that derivation: a fixed flux limit raises the luminosity threshold with the distance modulus, by 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$; the Zehavi slope below is 0.5–1.0 in bias per magnitude near LRG luminosities; their product, 0.13–0.27 in bias, is 6–11% of $b \approx 2.4$. The caveat is stronger than for BGS: the threshold-bias fit is an $r$-band relation measured at $z < 0.25$, applied here to a $z$-band fiber-magnitude selection at $z \approx 0.8$–1.1, outside its calibrated range, so the drift's direction and order of magnitude are what the argument establishes.

**White et al. [17] and Sailer et al. [18].** What was measured: cosmological constraints from the cross-correlation of the photometric LRG sample of [16] with CMB lensing from Planck, and in [18] also ACT DR6, in four tomographic bins over $0.4 < z < 1$. Method: angular cross- and auto-spectra with one free bias parameter per bin. Result: smooth models fit all bins. Implication here: these analyses neither require nor exclude percent-level bias structure between thin bins; they bound nothing below their bin width of about 0.15 (measured context).

### F.3 Bias versus luminosity threshold

**Zehavi et al. [20].** What was measured: the luminosity and color dependence of galaxy clustering in the SDSS DR7 main sample, about 700,000 galaxies at $z < 0.25$. Method: projected correlation functions of volume-limited luminosity-threshold samples, interpreted with HOD modeling. Result: the large-scale bias of a threshold sample is fit by $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L/L_*)^{1.12}$, with $L_*$ the characteristic luminosity of the $r$-band luminosity function and $\sigma_8$ the rms matter fluctuation in spheres of $8\,\mathrm{Mpc}/h$; the measured threshold biases rise from $1.40 \pm 0.03$ at $M_r < -21.0$ to $2.16 \pm 0.05$ at $M_r < -22.0$, a slope of 0.5–1.0 in bias per magnitude of threshold near LRG luminosities. Implication: this relation converts a selection-threshold drift into a bias drift; the LRG3 and BGS drifts of Appendix B.5 are derived through it, and Appendix E spells out the BGS conversion step by step.

### F.4 The bright galaxy survey

**Hahn et al. [23].** What was measured: the BGS target selection and its validation. Method: selection on Legacy Surveys DR9 imaging plus Survey Validation spectra. Result: BGS Bright is a magnitude-limited sample at $r < 19.5$ with redshift success above 95% across a broad range of galaxies; BGS Faint adds $19.5 < r < 20.175$ with a color selection. Implication: the fixed

$r < 19.5$ limit raises the luminosity threshold with redshift, which drives the derived 10–25% bias drift across the BGS group (measured limit, derived drift; the derivation is Appendix E).

**Sailer et al. [19].** What was measured: the growth of matter fluctuations at $z \lesssim 0.4$ from BGS imaging cross-correlated with Planck and ACT CMB lensing. Method: tomographic angular cross-spectra with a hybrid effective-field-theory model and per-bin bias nuisance parameters. Result: growth consistent with ΛCDM. Implication here: as with the LRG tomography, no resolution of bias structure between thin bins (measured context).

### F.5 The emission-line galaxies

**Raichoor et al. [25].** What was measured: the ELG target selection, $0.6 < z < 1.6$, and its spectroscopic efficiency. Method: a $g$-band fiber magnitude cut plus a $(g-r)$ versus $(r-z)$ color box; redshift reliability is a cut on the signal-to-noise of the [OII] 3727 Å doublet. Result: the [OII] signal-to-noise drops at the specific redshifts where the doublet lands on sky emission lines, especially at $z > 1.5$, and the catastrophic-failure fraction rises over $1.3 < z < 1.6$. Implication: these windows, each $\Delta z \approx 0.001$–$0.01$ wide, are the one short-separation selection mechanism found in any tracer; their bias amplitude is assumed, of order 1–2%, a judgment from the weak dependence of ELG bias on [OII] flux rather than a measured bound.

**Rocher et al. [26].** What was measured: the HOD of the DESI One-Percent ELG sample. Method: fits of AbacusSummit simulation halos to the measured small-scale clustering, in two redshift bins, $0.8 < z < 1.1$ and $1.1 < z < 1.6$. Result: every HOD parameter agrees within one sigma between the bins; the mean host halo mass is $\log_{10} M_h = 11.78^{+0.03}_{-0.04}$ at mean redshift 0.95 and $11.86 \pm 0.05$ at 1.325. Implication: the ELG bias shape is stable across ELG1 and most of ELG2 (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, with the slope assumed).

## G The accuracy test: the figure of bias

The ladder of Section 1 orders the linear-bias priors by how much information they keep. The accuracy test orders them by the other quantity that matters: whether each fit's quoted error still covers the truth when the true biases carry structure the prior cannot express. This appendix gives the method: the shift of a fit under a known drift, the two drift laws, the random truths built on them, and the test statistic $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of Eq. (6). The results under model A are in the Summary and in Appendix Q.

### G.1 The shift formula

Let $m$ be the expected data vector (the band powers), $\mathsf{C}_d$ its covariance, $\mathsf{D}$ the matrix of derivatives of $m$ with respect to the parameters (sans-serif, not the growth factor $D$), and $F = \mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm priors}$ the Fisher matrix of one fit, its priors included (Appendix N). Under model A the parameters also include the variant's higher-order biases, counterterms and stochastic amplitudes, marginalized like the cosmology (Appendix P.3). Now let the true biases carry a known drift the fit may not be able to express: the data vector moves by $\delta m$. To first order in

$\delta m$, the maximum of the posterior moves by

$$\delta p \;=\; F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\,\delta m, \tag{35}$$

the linear response of the fit. The derivation is one line: the score (the gradient of the log posterior) at the old maximum becomes $\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\delta m$ instead of zero, and the curvature that converts a score into a parameter displacement is $F$.

The quoted number compresses $\delta p$ into the plane that matters. With $\mathsf{C}_w$ the fit's marginalized $2\times 2$ covariance of $(w_0, w_a)$ and $\delta w$ the $(w_0, w_a)$ entries of $\delta p$, the shift is

$$\Delta_w \;=\; \sqrt{\delta w^{\mathsf{T}}\,\mathsf{C}_w^{-1}\,\delta w}\,: \tag{36}$$

the displacement of the dark-energy ellipse in units of its own size. $\Delta_w = 1$ means the fitted $(w_0, w_a)$ sits on the one-sigma contour of the unshifted forecast. The threshold drawn in the figures is $\Delta_w = 1/3$, a shift that widens the effective error by a few percent. Both the ellipse and the response use the same fit's $F$: a rigid model is punished twice, because it absorbs little of the drift and its ellipse is small.

The drift enters through the bin biases. With $\beta_i$ the bin amplitudes of Eq. (9) and $\delta_i$ the drift's fractional deviations, the data move by $\delta m = \sum_i (\partial m/\partial\beta_i)\, B_g\, \delta_i$, with each group's mean deviation removed because $B_g$ is free. Under model A the column $\partial m/\partial\beta_i$ is the chain-rule column of Eq. (90) divided by $D(z_i)$, so a drift of $b_1$ drags $b_2$ and $b_{s^2}$ along through their calibrations (Appendix P.4). For the growth-tied prior the response is evaluated in the growth-tied parameter basis; for the free prior the bias derivatives span the drift exactly, so the cosmology shift vanishes: the free biases absorb the drift, and the accuracy of that prior is bought entirely with the information the ladder shows it giving up.

### G.2 The drift laws

Two known drifts are injected, the two derived in these appendices. The conserved-population drift applies to every group: the deviation of Eq. (19), anchored at the group's mean redshift. It is the gentlest drift physics supplies, nearly a straight line of a few percent across each group. The BGS flux-limit drift applies to BGS alone: the convex deviation of Appendix E, a 22% span whose best line leaves a 2.4% rms curvature. Both come from `plots/make_theory_data.py`, the script behind the theory figures; only the shape of each drift matters, because the free amplitudes remove the level. Appendix Q.1 gives the shift of each drift alone under model A.

A known drift is only half of a bias history. The two drifts are the deviations these appendices can derive; a real sample adds residual structure whose shape nobody has measured. The rest of this appendix makes that structure random and asks each prior the sharper question: does the error ellipse you quote still cover the truth?

### G.3 Random truths

A random truth sets the bias of bin $i$ in group $g$ to

$$b_i \;=\; \frac{B_g}{D(z_i)}\left(1 + \delta_i^{\mathrm{drift}} + \varepsilon_i\right) \tag{37}$$

at the fiducial cosmology. The drift $\delta_i^{\mathrm{drift}}$ is the known part: the $r < 19.5$ flux-limit deviation in BGS and the conserved-population deviation of Eq. (19) in the other five groups. The residual

$\varepsilon_i$ is the random part: one draw from the Gaussian $\mathcal{N}(0, \Sigma)$ whose covariance is the kernel of one of the two stated priors, $\Sigma_{ij} = \sigma_g^2\, e^{-|z_i - z_j|/L}$ inside a group and zero across groups. Two families of 2000 truths are drawn: baseline scatter ($\sigma_g = 3\%$ for BGS, 1% for the other groups, $L = 0.2$) and pessimistic scatter ($\sigma_g = 3\%$ in every group, $L = 0.05$). Under model A the truth's $b_2$ and $b_{s^2}$ follow its $b_1$ through Eq. (5).

Every truth is fitted under every prior, each centered on the growth-tied shape $\delta = 0$. No fit is told which truth it faces: the baseline prior meets truths drawn from its own prior (the matched case below) and truths drawn from the pessimistic prior, and is judged on both.

The sampling costs one matrix product because the response is linear. Eq. (35) is linear in $\delta m$, and $\delta m$ is linear in the bias deviations, so the $(w_0, w_a)$ rows of the response collapse into one $2 \times n_{\rm bins}$ matrix $\mathsf{A}$ per prior, layout and higher-order variant, with $n_{\rm bins}$ the number of bins of the layout:

$$\delta w \;=\; \mathsf{A}\,\delta, \qquad \delta_i \;=\; \delta_i^{\rm drift} + \varepsilon_i. \tag{38}$$

The draws also check themselves against the closed form they must reproduce, $\langle \Delta_w^2 \rangle = \Delta_{\rm drift}^2 + \mathrm{tr}(\mathsf{C}_w^{-1} \mathsf{A} \Sigma \mathsf{A}^{\mathsf{T}})$, where $\langle \cdot \rangle$ averages over the truths, $\Delta_{\rm drift}$ is the shift of the drift alone and $\mathsf{C}_w$ is the fit's marginalized $(w_0, w_a)$ covariance.

One caveat bounds the method. The response is first order in $\delta$, and the band powers are quadratic in the bin biases, so a term of order $\delta^2$ is dropped: a few percent of the shift for the 1–3% residuals, and perhaps 10–20% of it for the 22% BGS drift. Appendix U lists it among the limitations.

Figure 3, in the Summary, draws the result under model A. The one subtlety is the matched prior: the baseline prior in its top panel and the pessimistic prior in its middle panel. Where the fit's prior equals the truth's scatter, the average shift is not a bias. That fit's ellipse is wider than the growth-tied model's precisely because it marginalizes over the scatter the truth now actually has: the fitted $(w_0, w_a)$ moves from draw to draw, and the ellipse was built to cover those moves. A correctly specified model therefore shows a nonzero average shift just by being correct, and the mean of $\Delta_w$ under the matched prior measures the share of the ellipse the bias scatter occupies, not an error of the model. Deciding whether the quoted ellipse is honest needs the statistic of the next subsection.

### G.4 The test statistic

The question the statistic answers is this: does the fit's quoted error bar honestly cover where the fit actually lands? A fit is correctly specified when its quoted ellipse covers the truth with the stated probability. The statistic that tests exactly that is the squared distance of the fitted $(w_0, w_a)$, written $\hat{w}$, from the truth $w_{\rm true}$ in units of the fit's ellipse, $\chi_w^2 = (\hat{w} - w_{\rm true})^{\mathsf{T}} \mathsf{C}_w^{-1} (\hat{w} - w_{\rm true})$, with every source of displacement counted: the drift, the bias scatter, and the data noise. Averaged over the bias history (the known drift plus a random residual) and the data noise, written $\langle \cdot \rangle$ as in Eq. (6), it is

$$\langle \chi_w^2 \rangle \;=\; \Delta_{\rm drift}^2 \;+\; \mathrm{tr}\Big(\mathsf{C}_w^{-1} \mathsf{A}\, \Sigma\, \mathsf{A}^{\mathsf{T}}\Big) \;+\; \mathrm{tr}\big(\mathsf{C}_w^{-1} \mathsf{N}_w\big)\,, \tag{39}$$

where $\mathsf{N}_w$ is the share of $\mathsf{C}_w$ contributed by the data noise and by the Gaussian priors other than the bias prior: the priors on the per-group velocity dispersions $\sigma_v$ and, under model A, those on the higher-order parameters. The cross terms average to zero, because the residual and the noise have zero mean. Eq. (39) is evaluated in closed form, not by sampling: it is the

exact average over infinitely many universes, each with its own noise realization and its own bias history. The 2000 residual draws of Appendix G.3 only draw the 16th–84th percentile bands of Figure 3, and those draws carry no data noise. The quoted number is

$$\sqrt{\langle \chi^2_w \rangle \,/\, 2}, \tag{40}$$

the case $n_{\rm dof} = 2$ of $\sqrt{\langle \chi^2 \rangle / n_{\rm dof}}$ in Eq. (6); the case $n_{\rm dof} = 1$, for $\Omega_m$ or $A_s$, is Appendix G.5.

Eq. (39) splits $\langle \chi^2_w \rangle$ into three sources, in order: the drift squared, the scatter the prior did not absorb, and the noise. The drift term $\Delta^2_{\rm drift}$ is the squared shift from the known drift alone, the same for every truth. The scatter term carries the truth's residual $\varepsilon$ into $(w_0, w_a)$ through the response matrix $\mathsf{A}$ of Eq. (38): it is the part of the scatter that the fit's bias parameters, held back by their prior, did not follow. The noise term counts the data noise and the other Gaussian priors.

An honest ellipse gives exactly 2, the number of parameters. Suppose the fit is correctly specified: no drift remains, and the fitted $(w_0, w_a)$ scatters around the truth with covariance exactly $\mathsf{C}_w$. The expected squared distance is then $\mathrm{tr}(\mathsf{C}_w^{-1}\mathsf{C}_w)$, the trace of the $2 \times 2$ identity matrix, so $\langle \chi^2_w \rangle = 2$: one unit for $w_0$ and one for $w_a$. A larger value means the fit lands, on average, farther from the truth than the reported ellipse allows.

Eq. (40) turns that excess into a length. Growing both axes of the ellipse by a factor $f$ replaces $\mathsf{C}_w$ by $f^2\,\mathsf{C}_w$ and divides $\chi^2_w$ by $f^2$, so $f = \sqrt{\langle \chi^2_w \rangle / 2}$ is the growth that brings the average back to 2. It is the factor by which the axes of the reported ellipse would have to grow to cover the truth as often as the ellipse claims: 1 for an honest ellipse, above 1 for an ellipse that is too small, below 1 for a conservative one. The factor matches the average squared distance. It matches the coverage exactly when the real scatter of the fitted $(w_0, w_a)$ has the shape and orientation of the reported ellipse.

**What the statistic says about coverage.** An honest error bar has a frequentist meaning. Draw many universes, each with its own noise realization from the data covariance $\mathsf{C}_d$ and its own bias history (the known drift plus one Gaussian residual from the truth's prior), and fit each one. If the error bar is honest, the fit's $1\sigma$ interval covers the truth in 68.3% of them.

For one parameter $p$ the link to the statistic is exact. In units of the fit's marginalized error $\sigma_p$, the closed-form average has the three terms of Eq. (39),

$$\langle \chi^2_p \rangle \;=\; \mu^2 \;+\; a^{\mathsf{T}} \Sigma\, a \;+\; \nu, \tag{41}$$

with $\mu$ the shift of the known drift alone, $a$ the row of the response that carries the fractional bias deviations into $p$ (the analog of $\mathsf{A}$ in Eq. (38)) divided by $\sigma_p$, $\Sigma$ the covariance of the truth's residual, and $\nu$ the noise term: the $(p,p)$ entry of $F^{-1}F_0F^{-1}$ divided by $\sigma_p^2$, where $F_0 = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm other}$ holds the data covariance and the Gaussian priors other than the bias prior.

Because the response is linear and both the noise and the residual are Gaussian, the pull $\zeta = (\hat{p} - p_{\rm true})/\sigma_p$ is exactly Gaussian, with mean $\mu$ and variance $v = \langle \chi^2_p \rangle - \mu^2$. The $1\sigma$ bar covers the truth when $|\zeta| < 1$, which happens in a fraction

$$P(|\zeta| < 1) \;=\; \Phi\!\left(\frac{1-\mu}{\sqrt{v}}\right) - \Phi\!\left(\frac{-1-\mu}{\sqrt{v}}\right) \tag{42}$$

of the universes, with $\Phi(x) = \frac{1}{2}\big[1 + \mathrm{erf}(x/\sqrt{2})\big]$ the cumulative distribution of a unit Gaussian. Honest means $\mu = 0$ and $v = 1$: then $\langle \chi^2_p \rangle = 1$ and the coverage is $2\Phi(1) - 1 = 68.3\%$.

When the prior absorbs the known drift ($\mu \approx 0$), the coverage depends on the statistic $s = \sqrt{\langle \chi_p^2 \rangle}$ alone,

$$P(|\zeta| < 1) \;=\; 2\,\Phi(1/s) - 1 : \tag{43}$$

68% at $s = 1$, 56% at 1.3, 50% at 1.5, 43% at 1.75, 38% at 2 and 31% at 2.5. For $s \geq 1$, $\mu = 0$ is the most generous split of $\langle \chi_p^2 \rangle$ between drift and variance, so otherwise Eq. (43) is an upper bound. For the baseline, pessimistic and free priors, which absorb the known drifts ($\mu \leq 0.3$), it is exact to 0.1%. For the growth-tied prior the drift dominates, and only Eq. (42) is right.

Under the pessimistic scatter and model A baseline, the fraction of universes whose $1\sigma$ bar covers the truth is, on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2, 44, 56, 29 and 68% in $\Omega_m$ and 47, 62, 42 and 68% in $A_s$ for the baseline prior; 66–68% for the pessimistic prior; and 68.3% for the free prior. For the growth-tied prior it is 3, 12, 6 and 3% in $\Omega_m$, where Eq. (43) would give 8, 17, 13 and 13%. For $(w_0, w_a)$ the same logic holds with the ellipse $\chi_w^2 < 2.30$, which contains 68.3% of a two-dimensional Gaussian; its exact coverage depends on how the excess of $\langle \chi_w^2 \rangle$ splits between the two directions, so only the statistic is quoted there.

**Worked case: the baseline-prior fit of a pessimistic truth.** The baseline prior states that each group's residual $\varepsilon$ around its free line (the free amplitude $B_g$ and tilt $s_g$ of Eq. (2)) is small and smooth. Its width, the rms $\sigma_g$ of one bin's residual, is 1% (3% for BGS), and its correlation length $L = 0.2$ keeps the residual coherent over $\Delta z \approx 0.2$. The pessimistic truth has residuals of $\sigma_g = 3\%$ with $L = 0.05$ in every group: 3% wiggles that change over $\Delta z \approx 0.05$.

The baseline prior charges a large penalty for bin biases that follow such wiggles, so the fit lets the biases follow them only in part. The band powers still carry the rest of each wiggle, and the fit moves the cosmological parameters, $(w_0, w_a)$ among them, to absorb what the biases do not.

The push on $(w_0, w_a)$ is not one fixed offset. The drift part is the same for every truth; the rest comes from the residual and changes from truth to truth, so over the 2000 truths the shift $\Delta_w$ of Eq. (36), in units of the baseline-prior ellipse, spreads over a range: the 16th–84th percentile bars of Figure 3. The observed universe is one draw: one of these displacements, reported with the baseline-prior ellipse.

Adding the noise term, Eq. (39) gives $\langle \chi_w^2 \rangle$, and Eq. (40) gives the factor $f$, which Figure 4 draws for this case under model A. To read a value, take $f = 2$ for illustration: the real scatter of the best-fit $(w_0, w_a)$ around the truth is then twice as wide per axis as the reported ellipse, and the contour the baseline prior reports as one sigma is the $0.5\sigma$ contour of the real scatter.

The figure of merit follows the same factor. $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is proportional to the inverse area of the ellipse; growing both axes by $f$ multiplies the area by $f^2$, so the FoM scales as $1/f^2$: at $f = 2$ the quoted FoM is four times too large. The rescaled FoM is not a number to quote: it rescales an ellipse whose center and shape still come from a prior narrower and smoother than the truth's scatter. If the pessimistic truth is possible, the fit to report is the pessimistic prior, which is matched to that truth.

The opposite mismatch errs on the conservative side. A truth with baseline scatter, fitted with the pessimistic prior, gives $f$ below 1 (the Summary gives the model A values): the real scatter is then narrower than the reported ellipse.

**The matched case.** The matched case lands on 1 by an identity, not by tuning. Collect all parameters in $\theta$, with every prior centered on the fiducial $\theta_{\rm fid}$, and let $\Delta\theta$ be the truth's departure from it: zero in the cosmology and in every parameter that carries another Gaussian prior, $B_g\,\delta_i$ in the amplitude $\beta_i$ of bin $i$. With $n$ the data noise, of covariance $\mathsf{C}_d$, the linear response of Eq. (35) puts the fit at $\hat\theta = \theta_{\rm fid} + F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}(\mathsf{D}\,\Delta\theta + n)$. Write $F = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm bias} + F_{\rm other}$, with $F_{\rm bias}$ the bias prior's precision (the projected precision of Eq. (25) in the amplitude block) and $F_{\rm other}$ the other Gaussian priors. Since $F_{\rm other}\,\Delta\theta = 0$, the fit's error is

$$\hat\theta - \theta_{\rm true} \;=\; F^{-1}\big(\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}n \;-\; F_{\rm bias}\,\Delta\theta\big)\,: \tag{44}$$

the noise, plus the part of the truth's departure that the bias prior held back. Treating each other Gaussian prior as one more independent measurement, the noise part has covariance $F^{-1}(\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm other})F^{-1}$.

In the matched case the fit's projected precision $\mathsf{P}_{\rm mem}$ is built from the same kernel covariance $\Sigma$ that the truth's scatter uses (taken in the units of the bin amplitudes, a factor $B_g^2$ per group), and every projection of that form obeys

$$\mathsf{P}_{\rm mem}\,\Sigma\,\mathsf{P}_{\rm mem} \;=\; \mathsf{P}_{\rm mem}. \tag{45}$$

The proof takes one line: with $\mathsf{Q} = \Sigma^{-1}$ and $\mathsf{M} = (\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U})^{-1}$, $\Sigma\,\mathsf{P}_{\rm mem} = \mathsf{I} - \mathsf{U}\mathsf{M}\mathsf{U}^{\mathsf{T}}\mathsf{Q}$, and $\mathsf{P}_{\rm mem}\mathsf{U} = 0$ removes the second term. So the scatter part of Eq. (44) has covariance $F^{-1}F_{\rm bias}F^{-1}$, the two covariances add up to $F^{-1}F\,F^{-1} = F^{-1}$, and their $(w_0, w_a)$ block is $\mathsf{C}_w$ itself: the scatter and noise terms of Eq. (39) sum to $\mathrm{tr}(\mathsf{C}_w^{-1}\mathsf{C}_w) = 2$. The widening the prior adds to the ellipse equals the error the scatter inflicts, and the rest of the ellipse is the noise the data carry anyway. The matched value of Eq. (40) is then $\sqrt{1 + \Delta^2_{\rm drift}/2}$, which departs from 1 only through the known drift; under model A the matched priors absorb both drifts below $\sigma/3$ (Appendix Q.1), so it stays within 3% of 1. Appendix R verifies the identity numerically.

**When a fit is conservative.** The same algebra gives a sufficient condition for an error bar that is not too small. Write $\mathsf{P}_{\rm mem}$ in its null-space form, Eq. (83), with the fit's kernel $\Sigma_{\rm fit}$, and let $\Sigma_{\rm true}$ be the truth's. If the truth's scatter is no larger than the fit's prior in every combination the prior charges, $\mathsf{N}^{\mathsf{T}}\Sigma_{\rm true}\mathsf{N} \preceq \mathsf{N}^{\mathsf{T}}\Sigma_{\rm fit}\mathsf{N}$ (where $X \preceq Y$ means $x^{\mathsf{T}}Xx \le x^{\mathsf{T}}Yx$ for every vector $x$), then $\mathsf{P}_{\rm mem}\Sigma_{\rm true}\mathsf{P}_{\rm mem} \preceq \mathsf{P}_{\rm mem}$, the error covariance is at most $F^{-1}$, and $\sqrt{\langle\chi^2_w\rangle/2} \le \sqrt{1 + \Delta^2_{\rm drift}/2}$: a prior wider than the truth wastes information, which the ladder prices, but does not understate the error. Whether a given pair of kernels meets the condition depends on the kernels and the bins; the Summary measures the model A cases directly. Scatter larger or rougher than the pessimistic prior would make the pessimistic prior's error bar too small in the same way as the worked case, and against that the ladder ends at the free prior: with $F_{\rm bias} = 0$, Eq. (44) leaves only the noise, so $\sqrt{\langle\chi^2_w\rangle/2} = 1$ for every truth expressible through the bin biases.

### G.5 The same test for $\Omega_m$ and $A_s$

The statistics above score the dark-energy pair $(w_0, w_a)$. The same known drifts and the same residual priors are scored for the matter density $\Omega_m$ and the primordial scalar amplitude $A_s$, each in units of its own marginalized error $\sigma_p$. For one parameter $p$ the squared distance of the fit from the truth is $\chi^2_p = (\hat p - p_{\rm true})^2/\sigma_p^2$, with $\hat p$ the fitted value; an honest error bar gives

$\langle\chi_p^2\rangle = 1$, so the statistic is $\sqrt{\langle\chi_p^2\rangle}$, the case $n_{\rm dof} = 1$ of Eq. (6): 1 for an honest error bar, above 1 the factor by which the bar is too small, and Eq. (42) converts it into the fraction of universes whose bar covers the truth. For a matched case the identity of Eq. (45) makes the scatter and noise terms sum to exactly 1, because the $(p,p)$ entry of $F^{-1}$ is $\sigma_p^2$; the statistic is then $\sqrt{1+\Delta^2_{{\rm drift},p}}$, with $\Delta_{{\rm drift},p} = |\delta p|/\sigma_p$ the shift of the drift alone, and departs from 1 only through the known drift.

$\Omega_m$ and $A_s$ can react more strongly to a bias error than $(w_0, w_a)$, because a bias error moves the clustering amplitude $b_i^2 A_s D^2(z_i)$ of bin $i$, which $A_s$ scales directly and $\Omega_m$ moves through the growth factor. The Summary gives the model A values (Figure 4).

# H The roughness scan

Scans of the bias prior are best read against one number, the roughness of the deviation between neighboring bins.

## H.1 One number controls smooth deviations

For deviations that change slowly from bin to bin, the exponential kernel of Eq. (8) does not constrain $\sigma_b$ and $L$ separately. The jump variance of Eq. (14), ${\rm Var}(\delta_i - \delta_j) \approx 2\,\sigma_b^2\,\Delta z/L$ for separations $\Delta z \ll L$, depends on them only through $\sigma_b^2/L$, so a scan in two parameters hides a one-parameter family: doubling $L$ while multiplying $\sigma_b$ by $\sqrt{2}$ admits the same smooth drift. The combination these appendices use is the roughness

$$x \;\equiv\; \sigma_b \sqrt{\frac{0.1}{L}} \;=\; \sigma_b \sqrt{\frac{2\times 0.05}{L}}, \tag{46}$$

the small-separation rms change of the deviation between two bins $\Delta z = 0.05$ apart, a rescaling of the roughness $\kappa = \sigma_b/\sqrt{L}$ of Eq. (15) into an rms with a definite separation. The stated priors sit at three values: the baseline prior at $x = 0.71\%$ for the non-BGS groups ($\sigma_b = 1\%$, $L = 0.2$) and $x = 2.1\%$ for BGS (3%, $L = 0.2$), the pessimistic prior at $x = 4.2\%$ (3%, $L = 0.05$).

The roughness corrects a statement made in Appendix B. The argument there reads a short $L$ as unphysical because no selection mechanism decorrelates neighboring bins that fast. That was too strong: $L = 0.05$ with the matching $\sigma_b$ admits the real smooth drifts perfectly well, since only $x$ matters for them. What a short $L$ also admits is jagged histories nobody has observed, and what it cannot do is admit a large drift and forbid the jagged histories at the same time.

# I The projected precision for two bins

Everything essential already happens with two bins. Call their amplitudes $\beta_1$ and $\beta_2$, with common fiducial value $B$, prior spread $s$ per bin, and correlation coefficient $r$ between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} \;=\; s^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{47}$$

(The inverse is checked by multiplying: the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal is $(1 - r^2)/(1 - r^2) = 1$.)

Now examine the prior penalty $\chi^2_{\text{prior}} = \Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$, where $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is the displacement from the fiducial. Two kinds of move matter.

**A uniform shift.** Take $\Delta\beta = a\,(1, 1)^{\mathsf{T}}$: both bins move up by the same amount $a$. Working out the quadratic form with Eq. (47),

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{a^2}{s^2(1 - r^2)}\,\big[\,(1 - r) + (1 - r)\,\big] \;=\; \frac{2a^2}{s^2\,(1 + r)} \;\neq\; 0. \tag{48}$$

The plain prior charges for a uniform shift. But a uniform shift of all $\beta_i$ is a change of $B_g$, a direction the growth-tied model leaves completely free. If the intermediate model kept this penalty, it would constrain $B_g$ and the comparison between the models would be unfair.

**A shape change.** Take $\Delta\beta = a\,(1, -1)^{\mathsf{T}}$: the bins move apart. The same algebra gives

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{2a^2}{s^2\,(1 - r)}. \tag{49}$$

This is the penalty the prior is *meant* to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and it relaxes to the independent value $2a^2/s^2$ as $r \to 0$.

A number makes the contrast concrete. At $r = 0.670$ (the worked value of Appendix A.2), the shape change costs $2a^2/(0.330\,s^2) = 6.1\,a^2/s^2$ while the uniform shift costs $2a^2/(1.670\,s^2) = 1.2\,a^2/s^2$. The correlated prior already charges five times more for the shape change than for the uniform shift; the projection below takes the uniform-shift charge all the way to zero.

**The projected precision.** The repair for the uniform shift is the projection $\mathsf{P}$ of Eq. (17). Appendix J explains where the formula comes from; here, compute it for two bins. Because the two rows of $\mathsf{C}^{-1}$ have the same sum, $\mathsf{C}^{-1}\mathbf{1} = \big[s^2(1 + r)\big]^{-1}(1, 1)^{\mathsf{T}}$ and $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} = 2/\big[s^2(1 + r)\big]$. Substituting into Eq. (17) and simplifying each entry,

$$\mathsf{P} \;=\; \frac{1}{2s^2(1 - r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1 - \beta_2)^2}{2s^2(1 - r)}. \tag{50}$$

The projected prior charges only for the *difference* of the two amplitudes. A uniform shift, $\beta_1 = \beta_2$, is now free, as in the growth-tied model; a shape change keeps exactly the penalty of Eq. (49).

The denominator is not an arbitrary constant. With $\mathrm{Var}(\beta_i) = s^2$ and $\mathrm{Cov}(\beta_1, \beta_2) = s^2 r$,

$$\mathrm{Var}(\beta_1 - \beta_2) \;=\; s^2 + s^2 - 2s^2 r \;=\; 2s^2(1 - r)\,: \tag{51}$$

the projected penalty is the squared jump between the bins divided by the prior variance of that jump. The whole construction reduces to one sentence: *the prior constrains the jump, not the level.*

# J Why the projection "sees" the uniform shift: three views

Equation (17) subtracts from $\mathsf{C}^{-1}$ a matrix built out of the single vector $v = \mathsf{C}^{-1}\mathbf{1}$: the subtracted piece is $v\,v^{\mathsf{T}}$ divided by the number $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$. (In this appendix $v$ is that vector, not the centered redshifts of the memory model.) A matrix built from one vector acts along one direction only, so the subtraction removes the prior's information about exactly one direction, the uniform shift, and leaves everything else alone. This appendix derives the formula three times, each derivation adding its own intuition. Throughout, a group has $n$ bins; $\Delta\beta$ and $\mathbf{1}$ are vectors with $n$ entries.

## J.1 View (a): subtracting the best-fitting uniform shift

Ask the following question: given a displacement $\Delta\beta$, how much of it can be explained as a pure change of the group amplitude? A change of $B_g$ by $c$ moves every $\beta_i$ by $c$, so the candidate explanation is $c\,\mathbf{1}$, and the unexplained part should be judged by the prior metric. Minimize over the scalar $c$ the quadratic form

$$\chi^2(c) \;=\; \big(\Delta\beta - c\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big). \tag{52}$$

Differentiating with respect to $c$ (the two cross terms are equal because $\mathsf{C}^{-1}$ is symmetric),

$$\frac{d\chi^2}{dc} \;=\; -2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big) \;=\; 0, \qquad \frac{d^2\chi^2}{dc^2} \;=\; 2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;>\; 0, \tag{53}$$

where the second derivative is positive because $\mathsf{C}^{-1}$ is positive definite, so the solution of the first equation is the minimum of $\chi^2(c)$. That solution is

$$a \;=\; \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{54}$$

For a diagonal $\mathsf{C}$ with variances $s_i^2$ this reduces to $a = \sum_i \Delta\beta_i/s_i^2 \big/ \sum_i 1/s_i^2$, the familiar inverse-variance weighted mean; Eq. (54) is its generalization to correlated errors: generalized least squares for a constant.

Keep the dimensions straight: $\Delta\beta$, $\mathbf{1}$ and the residual $w = \Delta\beta - a\,\mathbf{1}$ are vectors with $n$ entries ($w$ is a vector of bin amplitudes, unrelated to the dark-energy parameters); $a$ is a single number. The residuals $w$ satisfy one linear constraint (Eq. (56) below), so they span $n-1$ dimensions: one direction of the $n$-dimensional displacement space has been handed to the amplitude, and $n-1$ directions of genuine shape freedom remain.

## J.2 View (b): a Pythagorean split of the penalty

Write the displacement as the best uniform shift plus a residual,

$$\Delta\beta \;=\; a\,\mathbf{1} + w. \tag{55}$$

The first-order condition of Eq. (53) says precisely that

$$\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; 0\,: \tag{56}$$

the residual is orthogonal to the uniform direction *in the* $\mathsf{C}^{-1}$ *metric*. (With correlated errors, "orthogonal" must be measured with the same matrix that measures lengths.) Substituting the split into the full penalty, the cross term is $2a\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w = 0$, and

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;+\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{57}$$

This is the Pythagorean theorem in the prior metric: the squared length of the displacement is the squared length of its uniform piece plus the squared length of its shape piece.

Now compare with the projection. Inserting Eq. (54) into the subtracted term of Eq. (17) shows that the subtraction removes exactly $a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$, the uniform piece. What remains is

$$\Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{58}$$

Two consequences follow immediately. First, a pure uniform shift has $w = 0$, so its penalty vanishes. Equivalently $\mathsf{P}\,\mathbf{1} = 0$: the projected matrix sends the uniform direction to zero and charges nothing for it. Second, a pure shape change (a displacement that already satisfies Eq. (56)) has $a = 0$ and keeps its original penalty untouched. The projection removes one direction and alters nothing else.

### J.3 View (c): marginalizing a flat-prior amplitude

The third view explains why the projection, rather than some other uniform-shift repair, is the *correct* prior. Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{59}$$

with a flat prior on the scalar $B$. This is literally the intermediate model of Eq. (7), linearized around the fiducial. The prior density of $\beta$ is obtained by integrating $B$ out:

$$p(\beta) \;\propto\; \int dB\, \exp\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{60}$$

The exponent is quadratic in $B$. Writing $S = \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$ and completing the square,

$$\big(\beta - B\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^{\mathsf{T}}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta}{S}. \tag{61}$$

The $B$ integral is a Gaussian centered on $B_*$ with width $1/\sqrt{S}$; it contributes only a $\beta$-independent constant $\sqrt{2\pi/S}$. What survives in the exponent is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^{\mathsf{T}}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}, \tag{62}$$

exactly the projection of Eq. (17).

The same computation has a standard matrix name. Expanding the exponent of Eq. (60) in the joint variables $(\beta, B)$ gives a Gaussian with the $(n+1)\times(n+1)$ block precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{63}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must also be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is called the *Schur complement* over $B$, and it reproduces Eq. (17) term by term. So $\mathsf{P}$ is not a trick: it is precisely the statement "$B_g$ *is sampled with a flat prior and marginalized*", folded into one matrix.

Note also that $B_*$ in Eq. (61) is the same generalized-least-squares mean as $a$ in Eq. (54): the marginalization integrates over all amplitudes, and the best-fitting one dominates the integral.

### J.4 Checking views (b) and (c) on the two-bin case

For $n = 2$ the rows of $\mathsf{C}^{-1}$ in Eq. (47) have equal sums, so Eq. (54) gives the plain mean, $a = (\Delta\beta_1 + \Delta\beta_2)/2$, and the residual is the antisymmetric half-difference, $w = \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)\,(1,-1)^{\mathsf{T}}$. Its penalty, using the shape-change result of Eq. (49) with $a \to \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)$, is

$$w^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; \frac{2}{s^2(1-r)}\left(\frac{\Delta\beta_1 - \Delta\beta_2}{2}\right)^2 \;=\; \frac{(\Delta\beta_1 - \Delta\beta_2)^2}{2s^2(1-r)}, \tag{64}$$

which is the direct two-bin computation of Eq. (50). Views (a), (b) and (c) all land on the same matrix.

## K Many bins and several groups

### K.1 Three bins: total correlation versus direct links

A group now has $n$ bins at redshifts $z_1 < z_2 < \cdots < z_n$, unevenly spaced in general, with the prior covariance of Eq. (10),

$$\mathsf{C}_{ij} \;=\; s^2 \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{65}$$

This matrix is positive definite for any set of distinct redshifts (all its eigenvalues are positive), so the prior is a proper probability distribution and $\mathsf{C}^{-1}$ exists.

The structure of $\mathsf{C}^{-1}$ is best seen on three evenly spaced bins. With neighbor correlation $r = e^{-\Delta z/L}$, the outer pair sits two steps apart and carries $e^{-2\Delta z/L} = r^2$, so

$$\mathsf{C} \;=\; s^2\begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)}\begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{66}$$

One multiplication confirms the surprising entry. The $(1,3)$ entry of $\mathsf{C}^{-1}\mathsf{C}$ must be zero (it is an off-diagonal entry of the identity matrix), and indeed row 1 of $\mathsf{C}^{-1}$ times column 3 of $\mathsf{C}$ gives, dropping the prefactors,

$$1\cdot r^2 \;+\; (-r)\cdot r \;+\; 0\cdot 1 \;=\; r^2 - r^2 \;=\; 0. \tag{67}$$

Here is the point that looks paradoxical at first. Bins 1 and 3 *are* correlated ($\mathsf{C}$ contains $r^2$), yet their entry in $\mathsf{C}^{-1}$ is exactly zero. The two matrices answer different questions. $\mathsf{C}$ gives the *total* correlation: what $\delta_1$ tells you about $\delta_3$ when nothing else is known. $\mathsf{C}^{-1}$ gives

the *direct* correlation: what $\delta_1$ still tells you about $\delta_3$ once $\delta_2$ is already known, and that is nothing. All of the 1–3 correlation travels through bin 2.

Think of a chain of people passing a message. The first person's message reaches far down the chain, but the links are only between neighbors: nobody talks past the person standing next to them.

The exponential kernel is special in making this exact, because its correlations multiply along the chain:

$$e^{-(z_3-z_1)/L} \;=\; e^{-(z_3-z_2)/L}\, e^{-(z_2-z_1)/L}. \tag{68}$$

The long-distance correlation is the product of the step correlations, so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way, because the squared separations do not add up along the chain: $(z_3-z_1)^2 \neq (z_3-z_2)^2+(z_2-z_1)^2$. Its inverse is dense.

The same structure shows directly in the penalty. For $n$ evenly spaced bins, the quadratic form of the deviations $\delta_i$ (whose covariance is $\sigma_b^2\mathsf{R}$, with $\mathsf{R}$ the correlation matrix of Eq. (10)) regroups into a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r\,\delta_k\big)^2}{\sigma_b^2\,(1-r^2)}. \tag{69}$$

Each term compares one bin with $r$ times the bin before it: every bin is pulled only by its neighbor, and the long-range correlation comes from chaining the pulls. (Expanding the squares and collecting terms reproduces Eq. (66) for $n = 3$; Appendix L does it for general, uneven spacing.) This is the precise sense of "correlation only between adjacent bins": *directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this neighbors-only structure the Markov property.

### K.2 The tridiagonal inverse for $n$ bins

A matrix whose nonzero entries sit only on the diagonal and the two neighboring off-diagonals is called *tridiagonal*. The three-bin inverse of Eq. (66) is the smallest interesting example, and the pattern grows unchanged. For $n$ evenly spaced bins, where every adjacent pair has the same correlation $r = e^{-\Delta z/L}$ and $\mathsf{C}_{ij} = s^2\, r^{|i-j|}$,

$$\mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{70}$$

with diagonal $(1,\; 1+r^2,\; \ldots,\; 1+r^2,\; 1)$, off-diagonal entries $-r$, blanks zero, and the overall factor $1/\big[s^2(1-r^2)\big]$. The entries read off the chain sum of Eq. (69). An off-diagonal entry is the link between that pair of neighbors. An interior bin touches two links, one to the bin on its left and one to the bin on its right, and the two pulls together contribute $1 + r^2$; the first and last bins touch only one link each, hence the plain 1 in the corners.

With uneven spacing, each adjacent pair has its own step and therefore its own correlation,

$$r_k \;=\; \exp\!\left(-\,\frac{z_{k+1}-z_k}{L}\right), \qquad k = 1, \ldots, n-1, \tag{71}$$

and the inverse keeps the same shape with per-pair entries:

$$s^2\left(\mathsf{C}^{-1}\right)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\ \dfrac{1}{1-r_{k-1}^2} + \dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\ \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad s^2\left(\mathsf{C}^{-1}\right)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{72}$$

In words: the off-diagonal entry is the neighbor link of pair $k$, stronger when the bins are closer ($r_k$ nearer one); the diagonal entry of bin $k$ collects the links touching that bin, one term from the pair on its left and one from the pair on its right; the first and last bins carry one link only. Setting all $r_k = r$ turns the interior diagonal into $1/(1-r^2) + r^2/(1-r^2) = (1+r^2)/(1-r^2)$ and recovers Eq. (70). Appendix L derives Eq. (72) from the chain of neighbor links; the companion script `verify_prior.py` checks the three-bin, even and uneven forms numerically.

### K.3 The projection, group by group

Each group receives the projected precision of Eq. (17), built from its own $\mathsf{C}$. The properties derived in Appendix J carry over verbatim to $n$ bins. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. $\mathsf{P}$ is positive semidefinite, which means no displacement is *rewarded*: every move either costs $\chi^2$ or is free. And of the $n$ directions a group can move in, exactly one (the uniform shift) is free while the other $n-1$ are charged: in matrix language, $\mathsf{P}$ has rank $n-1$.

Several groups combine trivially. The prior of Eq. (8) is zero across groups, so the full prior covariance is block-diagonal, one block per group; and each group has its own free amplitude $B_g$, so each block is projected separately. The total prior precision is the block-diagonal matrix of the per-group projections.

### K.4 The two limits

The intermediate model contains both extremes.

As $L \to \infty$, every $r_k \to 1$, so the jump variance $2\sigma_b^2(1-r_k)$ of every adjacent pair goes to zero: the prior forces all deviations of a group to be equal. The only freedom left is their common value, a uniform shift, and that is the direction the projection leaves free. This is the growth-tied model, with one free $B_g$ per group, *whatever* the value of $\sigma_b$.

As $\sigma_b \to \infty$ at fixed $L$, the overall scale of $\mathsf{C}$ diverges, so $\mathsf{C}^{-1} \to 0$ and with it $\mathsf{P} \to 0$: the prior penalty vanishes and every bin's amplitude is free: the free model with flat priors.

Between the limits, $L$ and $\sigma_b$ interpolate continuously, which is what makes the model useful: the figure of merit can be traced as a function of how much bin-to-bin freedom the bias is granted.

## L The tridiagonal inverse from the chain of links

Because correlations multiply along the chain (Eq. (68)), the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$p(\delta_1, \ldots, \delta_n) \;=\; p(\delta_1)\, p(\delta_2 \mid \delta_1) \cdots p(\delta_n \mid \delta_{n-1}), \tag{73}$$

where each conditional is Gaussian with

$$\big\langle \delta_{k+1} \mid \delta_k \big\rangle \;=\; r_k\,\delta_k, \qquad \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) \;=\; \sigma_b^2\,\big(1-r_k^2\big). \tag{74}$$

Here $r_k = e^{-(z_{k+1}-z_k)/L}$ is, as in Appendices A.2 and K.2, the correlation coefficient of the adjacent bins $k$ and $k+1$: the kernel of Eq. (8) gives $\mathrm{Cov}(\delta_k,\delta_{k+1}) = \sigma_b^2\,r_k$ and $\mathrm{Var}(\delta_k) = \mathrm{Var}(\delta_{k+1}) = \sigma_b^2$.

The conditional mean and variance in Eq. (74) follow from nothing more than these covariances. Split $\delta_{k+1}$ into the part predicted by its neighbor and a leftover $e$,

$$\delta_{k+1} \;=\; a\,\delta_k + e, \tag{75}$$

and choose the number $a$ so that the leftover carries no correlation with $\delta_k$:

$$\mathrm{Cov}(e,\delta_k) \;=\; \mathrm{Cov}(\delta_{k+1},\delta_k) - a\,\mathrm{Var}(\delta_k) \;=\; \sigma_b^2\,r_k - a\,\sigma_b^2 \;=\; 0 \quad\Longrightarrow\quad a = r_k. \tag{76}$$

The leftover's variance follows by expanding $e = \delta_{k+1} - r_k\delta_k$:

$$\mathrm{Var}(e) \;=\; \mathrm{Var}(\delta_{k+1}) - 2r_k\,\mathrm{Cov}(\delta_{k+1},\delta_k) + r_k^2\,\mathrm{Var}(\delta_k) \;=\; \sigma_b^2 - 2r_k^2\sigma_b^2 + r_k^2\sigma_b^2 \;=\; \sigma_b^2\,\big(1-r_k^2\big). \tag{77}$$

For jointly Gaussian variables, uncorrelated means independent. So once $\delta_k$ is known, the only randomness left in $\delta_{k+1} = r_k\delta_k + e$ is $e$ itself: the conditional mean is $r_k\delta_k$ and the conditional variance is $\sigma_b^2(1-r_k^2)$, which is Eq. (74). In one line: knowing the neighbor explains the fraction $r_k^2$ of the bin's variance, and the fraction $1-r_k^2$ is new.

The factorization of Eq. (73) needs one more fact: conditioning on $\delta_k$ alone must be enough: the earlier bins must add nothing. This is again the multiplication of correlations along the chain. For any earlier bin $j \le k$,

$$\mathrm{Cov}(e,\delta_j) \;=\; \sigma_b^2\left[e^{-(z_{k+1}-z_j)/L} - r_k\,e^{-(z_k-z_j)/L}\right] \;=\; \sigma_b^2\,e^{-(z_k-z_j)/L}\left[e^{-(z_{k+1}-z_k)/L} - r_k\right] \;=\; 0, \tag{78}$$

because the first exponential factorizes through $z_k$ and the bracket is $r_k - r_k$. The leftover is therefore independent of every earlier bin, and the chain of one-step conditionals is the complete prior.

Taking $-2\ln p$ of Eq. (73) gives the penalty as the chain sum, the uneven-spacing version of Eq. (69):

$$\delta^{\mathsf{T}}\big(\sigma_b^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1}\frac{\big(\delta_{k+1} - r_k\,\delta_k\big)^2}{\sigma_b^2\,\big(1-r_k^2\big)}. \tag{79}$$

Expanding each square and collecting the coefficient of every product $\delta_i\delta_j$ reads off the precision matrix entry by entry. $\delta_1^2$ appears twice: in the first term (coefficient 1) and in the $k=1$ link as the predictor (coefficient $r_1^2/(1-r_1^2)$); the two add up to $1/(1-r_1^2)$. An interior $\delta_k^2$ appears in the $k-1$ link as the predicted bin ($1/(1-r_{k-1}^2)$) and in the $k$ link as the predictor ($r_k^2/(1-r_k^2)$). The last bin is only ever predicted: $1/(1-r_{n-1}^2)$. Each product $\delta_k\delta_{k+1}$ appears in its own link only, with coefficient $-2r_k/(1-r_k^2)$, i.e. a symmetric matrix entry $-r_k/(1-r_k^2)$. No product of non-neighbors ever appears, so the matrix is tridiagonal by construction. This is Eq. (72) in units of $\sigma_b^2$; replacing $\delta$ by $\Delta\beta = B_g\,\delta$ and $\sigma_b$ by $s = \sigma_b B_g$ gives the same matrix for the bin amplitudes.

# M The memory projection, step by step

Throughout, a group has $n$ bins, $\mathsf{C}$ is the kernel covariance $\mathsf{C}_{ij} = \sigma_b^2 e^{-|z_i - z_j|/L}$ of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ is the $n \times 2$ design matrix of Eq. (25), with $v$ the centered redshifts, and $\theta = (a, s_g)^{\mathsf{T}}$ collects a line's level and slope. (For the bin amplitudes $\beta$, every covariance carries a factor $B_g^2$ that cancels in each identity below.)

## M.1 The line fit, the split, and the marginalization

The three views of Appendix J generalize from the single column $\mathbf{1}$ to $\mathsf{U}$ without change of structure.

Minimizing the penalty of a displacement $\delta$ over the line, $\chi^2(\theta) = (\delta - \mathsf{U}\theta)^{\mathsf{T}}\mathsf{Q}\,(\delta - \mathsf{U}\theta)$, gives the normal equations of a generalized least-squares fit,

$$\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\,\hat{\theta} \;=\; \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta, \qquad \hat{\theta} = (\hat{a}, \hat{s}_g)^{\mathsf{T}}, \tag{80}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ a $2 \times 2$ matrix, positive definite whenever the group has at least two distinct redshifts (then $\mathbf{1}$ and $v$ are linearly independent and $\mathsf{U}$ has full column rank).

Writing $\delta = \mathsf{U}\hat{\theta} + w$, the normal equations say $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,w = 0$: the residual is orthogonal to both columns in the $\mathsf{Q}$ metric. The cross terms then vanish and the penalty splits,

$$\delta^{\mathsf{T}}\mathsf{Q}\,\delta \;=\; \hat{\theta}^{\mathsf{T}}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)\hat{\theta} \;+\; w^{\mathsf{T}}\mathsf{Q}\,w, \qquad \delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; w^{\mathsf{T}}\mathsf{Q}\,w\,: \tag{81}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $w = 0$, so $\mathsf{P}_{\mathrm{mem}}\mathsf{U} = 0$; the residuals obey two linear constraints, so $\mathsf{P}_{\mathrm{mem}}$ has rank $n - 2$; and $w^{\mathsf{T}}\mathsf{Q}\,w \geq 0$ makes it positive semidefinite.

Marginalization gives the same matrix. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0, \Lambda)$, $\Lambda = \mathrm{diag}(W^2, \tau^2)$ the prior variances of the level and the slope, the marginal covariance of $\delta$ is $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$, and the Woodbury identity inverts it:

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}. \tag{82}$$

As the prior widths grow, $\Lambda^{-1} \to 0$ and the right-hand side becomes $\mathsf{P}_{\mathrm{mem}}$. The deviation falls inverse-quadratically in the widths: expanding the middle inverse, the leading correction carries $\Lambda^{-1}$, so the deviation falls as $1/W^2$, a factor of one hundred per decade of $W$. The script of Appendix O verifies the fall.

## M.2 The null-space form

A second form of the projection makes its meaning plain. Let $\mathsf{N}$ be an $n \times (n-2)$ matrix whose columns span the combinations of bins that no straight line can produce: $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$, plain Euclidean orthogonality to $\mathbf{1}$ and to $v$. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{83}$$

The proof has three steps. Both sides annihilate $\mathsf{U}$: the left by construction, the right because $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Both sides map $\mathsf{CN}$ to $\mathsf{N}$: the right because $\mathsf{N}(\mathsf{N}^{\mathsf{T}}\mathsf{CN})^{-1}(\mathsf{N}^{\mathsf{T}}\mathsf{CN}) = \mathsf{N}$, the left because $\mathsf{QCN} = \mathsf{N}$ and the subtracted term meets $\mathsf{U}^{\mathsf{T}}\mathsf{N} = 0$. And the columns of $\mathsf{CN}$ and of $\mathsf{U}$ together span the whole space: if $\mathsf{CN}x + \mathsf{U}y = 0$, multiplying by $\mathsf{N}^{\mathsf{T}}$ gives $\mathsf{N}^{\mathsf{T}}\mathsf{CN}\,x = 0$, so $x = 0$ (the

middle matrix is positive definite), and then $y = 0$ ($\mathsf{U}$ has full column rank). Two matrices that agree on a spanning set are equal.

The meaning: since $\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta = (\mathsf{N}^{\mathsf{T}}\delta)^{\mathsf{T}}(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N})^{-1}(\mathsf{N}^{\mathsf{T}}\delta)$, the prior reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. The same identity with the single column $\mathsf{U} = \mathbf{1}$ gives the kernel-model projection of Eq. (17); there $\mathsf{N}$ spans the zero-sum combinations, the differences between bins.

### M.3 Worked examples

One or two bins are fully free. For $n = 2$ the matrix $\mathsf{U}$ is square and invertible (two distinct redshifts), so $(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U})^{-1} = \mathsf{U}^{-1}\mathsf{C}\,(\mathsf{U}^{\mathsf{T}})^{-1}$ and the subtracted term collapses to $\mathsf{Q}$: the projection vanishes. For $n = 1$ nothing is orthogonal to both columns, so $\mathsf{N}$ has no columns and the null-space form is the zero matrix: a line fixes a single value trivially.

Three evenly spaced bins show what survives. With spacing $\Delta$ and $r = e^{-\Delta/L}$, the centered redshifts are $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$, and the only direction orthogonal to $\mathbf{1}$ and $v$ is the discrete curvature, the second difference $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$. Its prior variance expands to

$$\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} \;=\; \sigma_b^2\,\big(6 - 8r + 2r^2\big) \;=\; 2\,\sigma_b^2\,(1-r)(3-r) \tag{84}$$

(the diagonal contributes $1+4+1 = 6$, the two nearest-neighbor pairs $-8r$, and the outer pair $+2r^2$), so the penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_b^2\,(1-r)(3-r)}. \tag{85}$$

The memory model charges only the curvature; it is blind to any straight-line drift. The kernel model's three-bin case (Appendix K.1) charges the two neighbor differences instead, so it taxes a drift as heavily as a zigzag of the same step size.

### M.4 The universal-kriging predictor

Predict bin $k+1$ from the first $k$ bins, $k \geq 2$, with flat priors on $\theta$. Write $\mathsf{C}_k$ for the past kernel block, $\mathsf{U}_k$ for the past rows of $\mathsf{U}$, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ for the next row, and $c$ for the vector of kernel covariances between bin $k+1$ and the past. Given $\theta$, the ordinary Gaussian conditional is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; c^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{86}$$

With a flat prior, the posterior of $\theta$ given the past is centered on the generalized-least-squares fit $\hat{\theta}$ of Eq. (80) applied to the past block. Equation (86) is linear in $\theta$, so averaging over the posterior replaces $\theta$ by $\hat{\theta}$.

For the exponential kernel the weights collapse. Correlations multiply along the chain (Eq. (68)), so every entry of $c$ is $r_k$ times the corresponding entry of $\mathsf{C}_k$'s last column, and

$$c^{\mathsf{T}}\mathsf{C}_k^{-1} \;=\; r_k\,e_k^{\mathsf{T}}, \tag{87}$$

with $e_k$ the $k$-th unit vector (one in the last past slot, zero elsewhere). Substituting both results into Eq. (86) gives Eq. (24): the fitted line evaluated at the new bin, plus one Markov step from the last residual.

### M.5 The finite-slope correlation

With a Gaussian slope of variance $\tau^2$ and the amplitude held in $B_g$, the deviations have the covariance of Eq. (22), so $\mathrm{Var}(\delta_i) = \tau^2 v_i^2 + \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \tau^2 v_i v_j + \sigma_b^2 e^{-|z_i - z_j|/L}$. For separations much larger than $L$ the kernel term is exponentially small, and dropping it gives the correlation coefficient of Eq. (23). When $\tau|v_i|$ and $\tau|v_j|$ both exceed $\sigma_b$, the $\sigma_b^2$ terms in the denominators become negligible and the coefficient tends to $v_i v_j / |v_i v_j| = \pm 1$: plus one for bins on the same side of the group center, minus one for opposite sides.

## N How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix:

$$F \;=\; F_{\mathrm{data}} \;+\; F_{\mathrm{priors}}, \tag{88}$$

where $F_{\mathrm{data}}$ is the usual data Fisher matrix and $F_{\mathrm{priors}}$ holds one block per prior. The projected precision of each group, $\mathsf{P}_{\mathrm{mem}}$ of Eq. (25) for the baseline and pessimistic priors, is added to the rows and columns of that group's bin amplitudes $\beta_i$. All other parameters (the cosmological parameters, the velocity dispersions with their own independent Gaussian priors, and the higher-order parameters of model A with theirs) are unchanged: the bias prior has no entries outside the bias block. The growth-tied prior is not a block of this kind; it sums each group's amplitude columns into one (Appendix P.4).

One consequence matters in practice. Bins that lost all their band powers to the scale cut carry no data, so their rows of $F_{\mathrm{data}}$ are zero. In the free model such a parameter is completely unconstrained and must be dropped from the matrix before inversion. With the correlated prior it stays: the prior ties the empty bin to its neighbors, which do carry data, so the empty bin follows the chain, and the matrix stays invertible as long as the group's data fix its free level and tilt. Appendix R describes the pseudo-inverse that guards the one system where a direction carries no information.

## O Numerical verification

The companion script `verify_prior.py`, in this folder, checks every identity used in these appendices, 23 checks in all, each printing one PASS or FAIL line with its measured deviation. For the kernel model: the two-bin closed forms of Eqs. (47) and (50); the three-bin inverse of Eq. (66), including its exactly zero $(1, 3)$ entry; the null direction, positive semidefiniteness and rank $n - 1$ of $\mathsf{P}$ on uneven redshifts; the tridiagonal inverses of Eqs. (70) and (72); the Pythagorean identity of Eq. (58) on random displacements; the equality of $\mathsf{P}$ with the Schur complement of the joint $(\beta, B)$ precision of Eq. (63); the decorrelation and variance of the neighbor-prediction leftover of Appendix L; the jump variance of Eq. (13) by Monte Carlo ($10^5$ draws); and the worked example of Appendix A.2 ($r = 0.670$, a jump of 8.1% at $\sigma_b = 10\%$, 8.9% in the Taylor form). For the physics of Appendix B: the closed-form bias evolution of Eq. (18) against a direct integration of the continuity equation, the conserved-population deviations and the neighbor rms values. For the memory model of Appendix C: the annihilation $\mathsf{P}_{\mathrm{mem}}\mathsf{U} = 0$ with rank $n - 2$ on uneven redshifts; the $1/W^2$ approach of the Woodbury inverse of Eq. (82) to the projection; the null-space identity of Eq. (83); the three-bin curvature penalty with its

closed-form variance $2\sigma_b^2(1-r)(3-r)$; the vanishing of the projection for one and two bins; the universal-kriging predictor of Eq. (24) against a direct Gaussian conditional; the end-to-end correlation of Eq. (23) tending to $\pm 1$; the straight-line fits of Appendix C.4, to the conserved-population drift of every group and to the BGS flux-limit drift; and the LRG3 magnitude-limit rate, 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$ with its 6–11% bias drift. Run in this folder with the cocoa Python, the script prints 23 PASS lines.

The theory figures (the bias shapes, the drift residuals and the prior draws) are drawn from data written by `plots/make_theory_data.py`, from the same flat ΛCDM growth factor and project fiducial amplitudes as the checks above; each figure folder follows the convention of Appendix P.4.

# P Implementation and the Fisher construction

## P.1 Model A in the likelihood

Model A runs in the project likelihood (`likelihood/desi_cl.py`) when its option `higher_order_bias` is set to 1. The example yamls `EXAMPLE_EVALUATE_MODEL_A1–A4.yaml` set it up on the four layouts, in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2.

Every model A parameter is per bin in the likelihood ($b_{1,i}$ and the offsets, $\alpha_{0,i}$ and $\mathrm{SN}_{0,i}$ of bin $i$); the grouping, one value per group or a value plus a tilt, is a statement of the parameter yamls, generated by `scripts/make_desi_params.py --model-a`, exactly as for the linear bias.

- **Quadratic and tidal bias.** The likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ with Eq. (5) and passes them to CosmoLike through the setter that takes the tidal bias per bin. The third-order bias is zero.

- **Counterterm.** CosmoLike's $k^2$ slot is $2\,b_1 b_K\,k^2 P_{\rm mm}$, with $b_K$ its per-bin derivative-bias coefficient and $k$ in units of $H_0/c$. The likelihood sets $b_K = \alpha_0/[2\,b_1 (c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}$Mpc. The counterterm therefore multiplies the nonlinear matter spectrum, CosmoLike's convention, where DESI's definition multiplies the linear one (subsection "The draft's data vector against CosmoLike" of `notes/higher_order_bias`).

- **Stochastic term.** A constant three-dimensional power projects exactly, without the Limber approximation, to an $\ell$-independent spectrum,

$$C_\ell^{ij}\big|_{\rm stoch} = \mathrm{SN}_0\, P_{ij}\, K_{ij}, \qquad K_{ij} = \int dz\, n_i(z)\, n_j(z)\, \frac{dz/d\chi}{\chi^2}, \tag{89}$$

  with $n_i$ the normalized Fingers-of-God-convolved redshift distribution of bin $i$ and $\chi$ the comoving distance. The unit $P_{ii} = 1/\bar{n}_i$ is fixed by the bin's surface density, so that the projection with the raw distribution gives the angular shot noise. A cross pair of two bins of one tracer sample takes the mean of their two powers; a pair of two samples takes none, because different galaxies share no stochastic term.

- **Projection.** The low-rank non-Limber engine projects each bin's full one-loop $P_{gg,i}$, counterterm included, and the adjacent cross pairs carry the bilinear one-loop pair spectrum (subsection "Gaps against model A" of `notes/higher_order_bias`). Appendix S shows why the counterterm needs this engine.

## P.2 The derivatives

The Fisher matrix needs the derivative of the data vector $m$ (the band powers) with respect to every parameter. The notebook `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb` builds them in three groups.

*Stencil derivatives.* The seven cosmological parameters ($10^9 A_s$, the spectral index $n_s$, the Hubble constant $H_0$, the baryon density $\Omega_b$, $\Omega_m$, $w_0$, $w_a$), the six group amplitudes $B_g$ and the velocity dispersions come from a five-point stencil of the model A likelihood, as in v1's notebook.

*Per-bin operator columns.* The four columns $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$ and $\partial m/\partial \alpha_{0,i}$ of every bin, each at fixed values of the other three, come from central differences through the likelihood's bias setter. A wrapper intercepts the per-bin biases the likelihood passes to CosmoLike and shifts one of them for a set of bins. The steps are 1% of $b_1$, 0.02 in $b_2$ and $b_{s^2}$, and $5\,h^{-2}\,\mathrm{Mpc}^2$ in $\alpha_0$. The set of bins is one color of a greedy coloring of the cross-pair graph: two bins joined by a measured cross pair never share a color. Each spectrum then involves at most one shifted bin, so one pair of evaluations gives the columns of every bin of that color. Every layout needs four colors, so each operator costs eight evaluations instead of two per bin.

*Stochastic columns.* The $\mathrm{SN}_0$ columns are the exact templates $P_{ij}K_{ij}$ of Eq. (89); the vector is linear in $\mathrm{SN}_0$.

Model A ties $b_2$ and $b_{s^2}$ to $b_1$, so the derivative with respect to a bin's linear bias is the chain-rule sum

$$\left.\frac{dm}{db_{1,i}}\right|_{\mathrm{A}} \;=\; \frac{\partial m}{\partial b_{1,i}} \;+\; b_2^{\mathrm{L}\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} \;-\; \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}, \tag{90}$$

with $b_2^{\mathrm{L}\prime}$ the slope of the Lazeyras fit. Summed over a group with weights $1/D(z_i)$, these columns must reproduce the stencil derivative of the amplitude $B_g$ (Appendix R).

## P.3 The variants and the Fisher matrices

The parameters are, in order: the cosmological parameters and the velocity dispersions (with v1's Gaussian priors), the variant's higher-order parameters, and one amplitude $\beta_i = b_{1,i}\,D(z_i)$ per bin, fiducial $\beta_i = B_g$ (Eq. (9)), as in v1. The column of $\beta_i$ is Eq. (90) divided by $D(z_i)$. The higher-order columns of group $g$ are sums over its bins:

$$\frac{\partial m}{\partial \Delta_{2,g}} = \sum_{i\in g} \frac{\partial m}{\partial b_{2,i}}, \qquad \frac{\partial m}{\partial t_{2,g}} = \sum_{i\in g} (\bar{z}_i - \bar{z}_g)\,\frac{\partial m}{\partial b_{2,i}}, \tag{91}$$

and likewise for $\Delta_s$ and $\alpha_0$, with $t_{2,g}$ the tilt of $\Delta_2$ in group $g$; the $\mathrm{SN}_0$ column sums the group's stochastic templates. The exact-calibration configuration of Appendix T has no higher-order column; model A baseline has the offset columns with their Gaussian priors; model A pessimistic adds the tilt columns, with flat priors, for every group of more than one bin.

The data Fisher matrix is $F_{\mathrm{data}} = \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D}$ on the kept entries, with $\mathsf{D}$ the matrix of the columns above and $\mathsf{C}$ the Gaussian covariance computed for 14,000 square degrees, its inverse rescaled to the survey's 12,300. The Gaussian priors add to its diagonal. The higher-order parameters stand in front of the per-bin amplitudes and are marginalized like the cosmology, so v1's ladder code acts unchanged on the amplitude block. Every Fisher matrix is inverted through a pseudo-inverse (Appendix R).

### P.4 The ladder, the response and the truths

- **Linear-bias priors.** Growth-tied sums the amplitude columns of each group into one; the baseline and pessimistic priors add the projected precision of the residual prior, Eq. (25), to the amplitude block (Appendices N and M); free adds nothing.
- **Response.** A truth that moves the data by $\delta m$ moves the best fit by $\delta p = F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\delta m$, to first order; the $(w_0, w_a)$ shift is $\Delta_w = (\delta w^{\mathsf{T}}\mathsf{C}_w^{-1}\delta w)^{1/2}$, with $\mathsf{C}_w$ the fit's marginalized $(w_0, w_a)$ covariance, and the $\Omega_m$ and $A_s$ shifts are $|\delta p|/\sigma_p$ (Appendix G.1).
- **Known drifts.** The two drift laws of Appendix G.2: a conserved population in every group and the $r < 19.5$ flux limit in BGS. Under model A a drift $\delta_i$ of the linear bias moves $b_2$ and $b_{s^2}$ as well: $\delta m = \sum_i (\partial m/\partial\beta_i)_{\mathrm{A}}\, \beta_i\, \delta_i$, with the model A column of Eq. (90).
- **Random truths and $\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}}$.** v1's 2000 draws per scatter family, with v1's seed and order (Appendix G.3); the statistic of Eq. (6), $\sqrt{\langle\chi^2_w\rangle/2}$ for $(w_0, w_a)$ and $\sqrt{\langle\chi^2_p\rangle}$ for one parameter $p$ (Appendices G.4 and G.5).
- **Calibration drift.** One deterministic truth that moves only the offsets: $\delta m = \sum_i [(\partial m/\partial b_{2,i})\,\Delta_{2,i} + (\partial m/\partial b_{s^2,i})\,\Delta_{s,i}]$, with $\Delta_{2,i} = (\bar{z}_i - \bar{z}_g)/(\bar{z}_{\max} - \bar{z}_{\min})$ and $\Delta_{s,i} = \Delta_{2,i}/2$, where $\bar{z}_{\max}$ and $\bar{z}_{\min}$ are the largest and smallest mean redshifts of the group's bins. The drift therefore changes by 1 in $\Delta_2$ and 0.5 in $\Delta_s$ from the group's first bin to its last, has zero mean over the group's bins, and vanishes in one-bin groups.

Every figure lives in its own folder under `plots/`, named after the figure, with its data (`data.json`, written by the notebook), one Python script that draws it, and the PDF the note includes. Each figure redraws from its folder alone.

## Q The known drifts, the test statistic and the calibration drift

Three figures carry the results of this appendix: Figure 9 (the known drifts), Figure 4 in the Summary ($\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}}$) and Figure 10 (the calibration drift). The full numbers, every prior, both variants and both truths, are in each figure's `data.json` (`plots/drift_shifts`, `plots/goodness_of_fit`, `plots/calibration_drift`) and in the printed tables of `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb`. All use $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$, with each bin cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask; values below $10^{-9}$ are numerical zeros.

### Q.1 The known drifts

Figure 9 draws the shift of each known drift alone. The free prior absorbs both drifts exactly. Two readings. The growth-tied prior stays excluded: the conserved-population drift moves its $(w_0, w_a)$ by 7.9–11$\sigma$ under model A baseline and 2.5–3.4$\sigma$ under model A pessimistic. The baseline and pessimistic priors absorb both drifts below $\sigma/3$ in every score; the largest entry, 0.32$\sigma$, is the BGS flux limit on $\Delta z = 0.05$ (pessimistic prior, model A baseline), a layout that resolves BGS into only six bins.

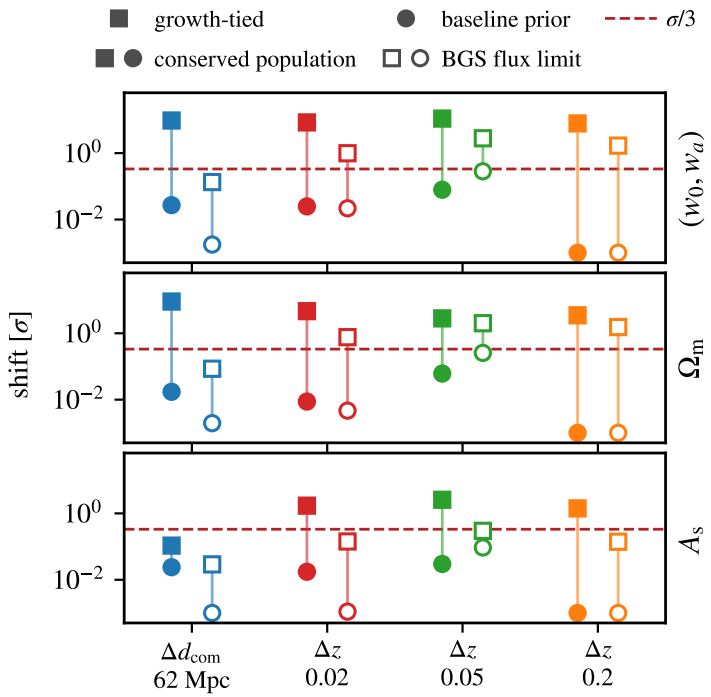



Figure 9: The shift of a model A baseline fit under each known drift: $(w_0, w_a)$ in units of the fit's ellipse, $\Omega_m$ and $A_s$ in their own marginalized $\sigma$, per layout. Squares: the growth-tied prior; circles: the baseline prior. Filled: the conserved-population drift; open: the BGS $r < 19.5$ flux limit. Dashed red line: $\sigma/3$. Points at $10^{-3}$: values below it. Omitted: the pessimistic prior, at most $0.318\sigma$ in $(w_0, w_a)$, 0.304 in $\Omega_m$ and 0.182 in $A_s$; the free prior, about $10^{-11}$; model A pessimistic, whose growth-tied fit moves by 2.5–3.4$\sigma$ in $(w_0, w_a)$ under the conserved drift and whose baseline and pessimistic priors stay below $0.103\sigma$ in every score. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

## Q.2 The test statistic

Figure 4, in the Summary, draws $\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}}$ of Eq. (6) under model A baseline and pessimistic scatter; its `data.json` holds every prior, both variants and both truth families. A value of 1 means the error covers the truth as often as it claims; below 1 the error is conservative. The free prior absorbs every random truth exactly.

Two readings beyond the Summary. The growth-tied fit's $A_s$ error is the least damaged of its scores ($\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}} = 1.1$ on $\Delta d_{\mathrm{com}} = 62$ Mpc under baseline scatter), but its $(w_0, w_a)$ error, at least 2.0 times too small, excludes it on every layout. And under model A pessimistic the growth-tied fit's $(w_0, w_a)$ value falls to 2.0–4.4, against 5.6–10.3 under model A baseline. A likely reason: the flat-prior tilts of the higher-order offsets can follow part of a linear-bias drift. That narrows the gap without making the growth-tied fit honest.

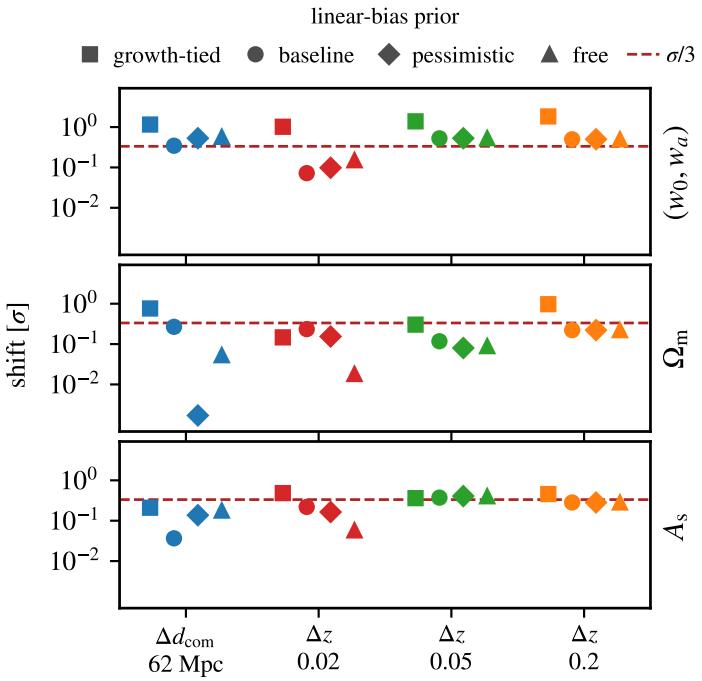



Figure 10: The shift of a model A baseline fit under the calibration drift ($\Delta_2$ from about $-0.5$ to $+0.5$ and $\Delta_s$ from about $-0.25$ to $+0.25$ across each group): $(w_0, w_a)$, $\Omega_m$ and $A_s$, per layout. Markers: the four linear-bias priors (square growth-tied, circle baseline, diamond pessimistic, triangle free). Dashed red line: $\sigma/3$. Omitted: model A pessimistic, which absorbs the drift exactly (every shift at most $1.6 \times 10^{-12}$). $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

## Q.3 The calibration drift

Figure 10 draws the shifts of the calibration-drift truth under model A baseline. Model A pessimistic absorbs the drift under every prior and on every layout (every shift at most $1.6 \times 10^{-12}$) and is omitted. Under model A baseline the baseline, pessimistic and free priors exceed $\sigma/3$ in $(w_0, w_a)$ on three layouts, and in $A_s$ on $\Delta z = 0.05$ (0.37–0.41$\sigma$).

# R Numerical checks

- **The chain rule.** For every group, the per-bin columns of Eq. (90), summed with the weights $1/D(z_i)$, reproduce the stencil derivative of the group amplitude $B_g$ to $9.8 \times 10^{-8}$, $1.2 \times 10^{-7}$, $4.0 \times 10^{-8}$ and $1.3 \times 10^{-8}$ of the stencil column's largest entry (layout order). The two independent routes, operator columns through the bias setter and the full likelihood through the stencil, agree.

- **The matched-prior identity.** When a fit's prior equals the truth's scatter, the projection identity $\mathsf{P\Sigma P} = \mathsf{P}$ of Appendix G.4 makes the scatter and noise terms of $\langle \chi^2_w \rangle$ sum to exactly 2. Over every layout and variant, $\max |\langle \chi^2_w \rangle - \Delta^2_{\rm drift} - 2| = 4.0 \times 10^{-10}$.

- **Exact recovery.** The free prior must absorb any truth of the linear bias. Its largest mean $(w_0, w_a)$ shift over the random truths is $2.6 \times 10^{-11}$.

- **One direction without information.** On $\Delta d_{\rm com} = 62$ Mpc at this cut only 7 of the 18 BGS bins keep any band. Under model A pessimistic on the free prior, the flat-prior BGS tilts of $\Delta_2$, $\Delta_s$ and $\alpha_0$ then leave one combination of parameters with no information. A plain matrix inverse amplified round-off along it into a spurious $2\sigma$ shift. The notebook therefore inverts every Fisher matrix through a pseudo-inverse: it scales the matrix to unit diagonal, diagonalizes it, and drops the directions whose eigenvalue is below $10^{-12}$ of the largest. Exactly one of the 48 systems (four layouts, four linear-bias priors, and three higher-order configurations: model A baseline, model A pessimistic and the exact calibrations of Appendix T) drops one direction, and that direction's largest component on $(w_0, w_a)$, $\Omega_m$ and $A_s$ is $5 \times 10^{-12}$: a pure nuisance combination. The FoMs are unchanged.

- **The $\alpha_0$ step.** At the negative stencil point, $\alpha_0 = -5\,h^{-2}\,{\rm Mpc}^2$, the factor $b_1^2 + \alpha_0 k^2$ of the linear and counterterm pieces turns negative above $k \approx 0.6\,h/{\rm Mpc}$, where the low-rank engine floors $\sqrt{P_{gg}}$. On $\Delta z = 0.05$, a step of 5 against a step of 1 changes the $\alpha_0$ columns by at most $3.5 \times 10^{-4}$ of each column's largest entry. Times a one-sigma $\alpha_0$ ($88.9\,h^{-2}\,{\rm Mpc}^2$), the difference is worth $\Delta\chi^2 = 2.7 \times 10^{-4}$ in the worst bin and $1.0 \times 10^{-3}$ with every bin shifted at once.

- **Reproducibility.** A recomputation of the $\Delta z = 0.05$ derivatives is bitwise identical to the cache.

- **The loop templates.** With $b_3 = 0$, CosmoLike's C-FAST-PT loop templates and Python FAST-PT agree to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ on the $\Delta d_{\rm com} = 62$ Mpc vector, pairs included (subsection "Gaps against model A" of `notes/higher_order_bias`).

- **The same data as v1.** The kept band powers are read from v1's cut cache at $k_{\rm max} = 0.075\,h/{\rm Mpc}$, and the v1 FoMs of Appendix T from v1's figure data.

# S The counterterm: engines, scale cut and sign

Three properties of the counterterm $\alpha_0 k^2 P_{\rm mm}$ were measured on the $\Delta z = 0.05$ layout at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ ($z = 0$ value). The first is settled and fixes how the forecast is computed; the other two are open.

## S.1 The response needs the low-rank engine

CosmoLike has two non-Limber engines for $C_\ell^{gg}$. FKEM, the FFTLog method of Fang et al. [8], projects the linear term exactly and moves everything nonlinear, the counterterm included, through its Limber difference, so its one-loop content is Limber. The low-rank engine factorizes each bin's full one-loop $P_{gg}$, less the white-noise constant of the loops (which Limber projects exactly), and projects it without the Limber approximation.

For the counterterm the difference is large. The low-rank $\alpha_0$ response, $\partial C_\ell/\partial\alpha_0$, is 7.3 times the Limber one at $\ell = 20$, 1.8 times at $\ell = 50$, 1.26 times at $\ell = 100$ and about 1.05 times at $\ell = 200$–$400$ (median over the kept auto spectra). The reason is the radial modes of thin bins. Limber evaluates the power at the transverse wavenumber $k = (\ell + 1/2)/\chi$ only. A thin bin with sharp edges also passes radial modes of higher $k$, and the $k^2$ weight of the counterterm amplifies them; at low $\ell$, where the transverse wavenumber is small, they dominate the response.

The low-rank response is converged: raising `nonlimber_accuracyboost` from 32 to 128 changes it by $\Delta\chi^2 = 6 \times 10^{-6}$, the rank from 5 to 10 by $5 \times 10^{-14}$, and the overall accuracy boost from 1 to 2 by $8\times10^{-4}$. For a one-sigma $\alpha_0$ the two engines' responses differ by $\Delta\chi^2 \approx 90$, while at $\alpha_0 = 0$ they agree on the full vector to $\Delta\chi^2 = 3.6$. The forecast uses the low-rank engine.

## S.2 Open: the counterterm's share beyond the cut

Apply the ultraviolet fraction of Eq. (1) to the $\alpha_0$ template $\partial C_\ell/\partial\alpha_0$ instead of the band power. On the kept bands it is a median 0.076 and a maximum 0.14, and every kept band is above 0.05. For the fiducial model A vector the same statistic is a median 0.017 and a maximum 0.052.

The cut was built on the linear model, which keeps every band below 5%. The counterterm template draws more of its signal from $k > k_{\rm max}/D(\bar{z}_i)$, the scales the cut is meant to exclude: part of what the forecast learns about $\alpha_0$ comes from there. Whether to rebuild the cut on the one-loop model, its templates included, is an open decision for the PI.

## S.3 Open, for chains: negative power at negative $\alpha_0$

The linear and counterterm pieces of Eq. (4) combine to $(b_1^2 + \alpha_0 k^2)\, P_{\rm mm}$, which is negative for $\alpha_0 < 0$ above $k \approx b_1/\sqrt{|\alpha_0|}$. At $\alpha_0 = -88.9\, h^{-2}\,{\rm Mpc}^2$, one prior width below zero, $P_{gg} - P_\infty$ (the spectrum the low-rank engine factorizes, with $P_\infty$ the white-noise constant of the loops) turns negative above $k \approx 0.14\, h/{\rm Mpc}$, and the engine floors 47% of its grid. Its vector then curves in $\alpha_0$: over $\pm 88.9\, h^{-2}\,{\rm Mpc}^2$ the departure from a straight line is worth $\chi^2 = 18.9$, while the FKEM vector is exactly linear ($2 \times 10^{-24}$).

This is harmless for the Fisher forecast, which is evaluated at $\alpha_0 = 0$ with small steps (Appendix R). Chains that explore $\alpha_0$ far below zero need one of the remedies the higher-order note gives for its white-noise constant (subsection "The white-noise limit of the renormalized loops"): a taper that returns $P_{gg}$ to $b_1^2 P_{\rm mm}$ above the scale where one-loop perturbation theory is trusted, or a positivity prior that rejects points with $P_{gg} < 0$ below a chosen $k$.

# T Comparison with linear bias and with exact calibrations

Neither comparison in this appendix is a result of this note. The linear-bias FoMs of v1 come from a model that `notes/linear_bias_kmax` shows is already biased at this cut, and the exact-calibration FoMs assume that $b_2$ and $b_{s^2}$ are known exactly, which no forecast can assume. They appear here only to locate model A's FoMs. Both sets of FoMs are in `plots/baseline_ladder/data.json` (the v1 values and those of model A calibrated) and in the notebook's printed ladder table.

**Linear bias.** Against v1's linear bias, model A baseline keeps about 84, 86, 71 and 44% of v1's FoM$(w_0, w_a)$ on the baseline prior, and model A pessimistic about 75, 73, 53 and 30% on the pessimistic prior (layouts in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2). The loss grows as the bins widen. A likely reason: on the wide layouts a group holds few bins, so one offset per group is close to one offset per bin, and the new parameters compete with each bin's linear bias for the same band powers. v1's FoMs remain the linear-bias reference.

**Exact calibrations.** With the calibrations taken as exact, the FoM exceeds v1's under every prior and on every layout: on the baseline prior it is about 1.3 times v1's on $\Delta d_{\rm com} = 62$ Mpc and about 1.9 times on $\Delta z = 0.05$. The loop terms are then not free: with $b_{2,i}$ and $b_{s^2,i}$ fixed by the calibrations at $b_{1,i}$, and $\alpha_0 = {\rm SN}_0 = 0$, the one-loop correction is a known function of $b_{1,i}$ and of the matter power spectrum. The loop terms grow as the fourth power of the growth factor and the linear term as $b_1^2$ times its square, so the size of the correction relative to the linear term carries information on the amplitude of the matter fluctuations that a linear-bias model does not have.

That information is only as good as the calibrations, and it is why this configuration is not a forecast. The offsets of model A baseline, with widths set by the known scatter of the calibrations, and its $\alpha_0$ and ${\rm SN}_0$ remove it, and model A baseline sits below v1 under every prior and on every layout. The calibration drift of the Summary shows what trusting the calibrations costs: with exact calibrations it shifts $(w_0, w_a)$ by 1.4–3.5$\sigma$ under the growth-tied prior and by 0.19–1.01$\sigma$ under the baseline, pessimistic and free priors. The notebook also prints this configuration's known-drift shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$; they are not repeated here.

# U Limitations

- **The covariance.** It is Gaussian and carries linear bias only. It also carries v1's mismatch: the covariance adapter (`_lens_bias`) uses each group's raw bias while the data vector uses $B_g/D(\bar{z}_i)$, which understates the high-redshift variances and inflates every absolute FoM by about 2.4–3.2. The comparisons between priors and variants hold the covariance fixed.

- **Linear response.** The shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ are first order in the truth's departure from the fiducial, as in v1. The dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift (Appendix G.3).

- **Theory truths, not mocks.** The project is a theory paper. The truths are the derived drifts, Gaussian residuals of the two stated sizes, and one linear drift of the

calibrations. Residuals rougher than the pessimistic prior, and calibrations that drift nonlinearly within a group, are untested.

- **The counterterm.** Its template's share beyond the cut and the negative power at negative $\alpha_0$ are open (Appendix S).
- **Real-space loops.** Model A adds the real-space one-loop terms to linear Kaiser distortions. Model B, at $k_{\rm max} \approx 0.18\,h/{\rm Mpc}$ ($z = 0$) with the redshift-space one-loop terms, the radial counterterm and the scale-dependent stochastic term, is not done.
- **Third-order bias.** $b_3 = 0$, as in DESI's baseline; CosmoLike's $b_3$ template is a fixed table, acceptable only while $b_3 = 0$ (`notes/higher_order_bias`).

# References


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